BULACAN STATE UNIVERSITY

COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY

IT303: SYSTEMS ANALYSIS AND DESIGN PROJECT DOCUMENTATION

Project Title	Dugo +
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TABLE OF CONTENTS

CHAPTER 1

PROJECT CONTEXT

1.1 Introduction

The Philippine Red Cross is the country's primary provider of voluntary blood services. Through its National Blood Services Program, it collects, tests, processes, stores, and distributes safe blood and blood products to hospitals and patients nationwide, and as of early 2023 it operated 31 blood centers and 71 blood collection units (Philippine Information Agency, 2023). Its work follows the National Blood Services Act (Republic Act No. 7719, 1994), which promotes voluntary, non-remunerated blood donation. Beyond regular donation drives, it runs awareness campaigns and works with government agencies, local government units, schools, private organizations, and healthcare providers to keep the blood supply sufficient for routine medical needs and emergencies, including surgeries, childbirth, trauma care, and cancer treatment.

Hospitals, blood banks, and local health offices still search for blood donors mostly by hand. When a recipient needs blood, hospital staff determine the required blood type and go through donor records that are commonly kept in paper files or spreadsheets (Eder, 2018). If no suitable donor is available for an emergency transfusion, staff phone or text previous donors. Hospitals, relatives, and community organizations sometimes post urgent requests on social media platforms such as Facebook to reach potential donors (Philippine Information Agency, 2025). Blood bank personnel and hospital coordinators then verify each donor's eligibility, confirm availability, and arrange the donation schedule. The paperwork involved consists of donor registration forms, Recipient blood request forms, blood inventory records, and donation history logs, and the tools are generally limited to spreadsheets, mobile phones, SMS, email, and social media. With donor information spread across multiple records and coordination done manually, identifying and contacting compatible donors can take considerable time, particularly when a transfusion is needed immediately.

This manual process slows response time, puts recipients at risk, and cannot scale under the pressure of a medical emergency. Because no single centralized public system exists, hospital staff search paper logs and spreadsheets on their own while relatives rely on personal contact lists and social media appeals, and confirming a donor's eligibility and availability takes considerable time. When hospitals cannot secure enough supply, relatives are sometimes asked to source blood themselves, a practice the Department of Health has flagged as risky: relatives have reportedly had to transport and store the blood on their own, in some cases in home freezers, where it spoils (Philippine Information Agency, 2025). Blood handled outside the standard collection-to-transfusion chain also bypasses the typing and cross-matching safeguards meant to prevent mismatched transfusions, which raises the risk of a transfusion reaction (A.D.A.M. Medical Encyclopedia, 2025). Part of the shortage is a donor-pool problem as well. The Department of Health says at least one percent of the population needs to donate to keep the supply sufficient (SunStar Cebu, 2026), yet a study of Metro Manila adults describes donation practice in the Philippines as low despite recruitment efforts (Mappala et al., 2023). Without a way to track and re-engage willing donors, the pool stays inactive and every new request starts the search from zero.

Donor records are scattered across non-centralized spreadsheets and paper logs, which makes monthly reporting and data retrieval slow and error-prone. The problem is well documented: the Philippine Red Cross built its own Online Blood Bank Management Information System to address it (Eder, 2018). The disorganization creates a data privacy concern as well. Donor information typically includes blood type, contact details, and health history, which counts as sensitive personal

information under the Data Privacy Act of 2012 (Republic Act No. 10173) and its health-sector implementing guidelines (Department of Health, 2020). Both require such data to be securely stored, access-controlled, and processed only with proper consent (Respicio & Co., 2025), a standard that paper logs and informal spreadsheets are not designed to meet.

The proposed system addresses these gaps by digitizing donor records into a centralized, searchable database of structured donor profiles that include blood type, current location, and donation history, so records can be retrieved instantly. It automates the search for compatible donors and sends and in app and email match alerts, removing the delay of manual coordination so that recipients receive aid sooner. The donor-facing mobile web app and the web platform for hospital, blood bank, and administrative users share a single workflow, from request submission to donor matching to confirmed donation, so hospitals, blood banks, and donors all work from the same real-time data. To keep that workflow transparent, the system records key transactions (requests, matches, confirmed donations, and reward points) in a blockchain transaction log. Each record carries a cryptographic hash of the record before it, so tampering with an earlier record can be detected. Existing blood donor management systems have been criticized for lacking a reliable audit trail and traceability, which blockchain-based designs have been proposed to address (Sadri et al., 2021). The hash-chained log applies the same chaining idea in a lighter form and gives hospitals, blood banks, and administrators a verifiable record of what happened and when.

Dugo+ is a Transaction Processing System designed to handle time-critical, high-volume donor-matching transactions accurately and immediately. It targets three friction points of the current manual process: slow donor identification, delayed treatment, and an inactive donor pool. The system serves four types of users: blood donors; Recipients or their family members who need blood; hospital or blood bank staff who coordinate donations; and system administrators who manage user accounts, system records, and the audit trail. Donors register and keep a profile containing their blood type, location, and donation history. When a hospital submits an urgent blood request, or a recipient's relative submits one that hospital staff then verify, the system automatically searches the donor records for people who match in blood type, are eligible to donate, and are located nearby. Once its GPS-based matching engine finds a match, Dugo+ immediately sends an app and email notification to both the donor and the requesting hospital. A matched donor who agrees to donate does so through the hospital or blood bank's standard collection, screening, and testing process, so Dugo+ speeds up the search for donors without bypassing those safeguards. Together, these functions aim to make donor coordination faster and more dependable.

1.2 Objectives

General Objective

To develop and design a real-time blood donor matching and emergency request system, composed of a mobile web application and a web platform, for hospitals and blood banks, including those of the Philippine Red Cross.

Specific Objectives

No.	Specific Objective
1	To Develop a centralized donor database that stores donor profiles and donation history

2	To Develop a mobile web application for donors and Recipients
3	To Implement a notification system for emergency blood donation requests
4	To Develop a location-based compatible donor matching engine that searches within a tiered radius of an incoming blood request.
5	To Evaluate the system using black box testing,

1.3 Scope

1.4 Limitations

Dugo+ coordinates donors and requests; it does not replace medical judgment. Its eligibility checks rely on recorded donation history and on what donors report about themselves, so final screening, typing, cross-matching, and infectious disease testing remain with licensed blood bank personnel. The system also does not cover blood collection, transport, storage, or transfusion, and it cannot guarantee that a matched donor will respond or that blood will be available in time. Donation stays voluntary. Requests from recipients or relatives are verified by hospital staff before donors are alerted, so verification depends on staff being available and can add delay, and the system cannot itself tell a genuine request from a false one.

Matching quality depends on the location data behind it. Smartphone GPS is typically accurate to within about 4.9 meters under open sky, but accuracy worsens near buildings, bridges, and trees (GPS.gov, 2022). Donors must also allow location access and keep their profile updated. Email delivery depend on third-party gateways and network conditions, so the response-time target applies to when the system sends notifications, not to when they arrive, and the system needs an internet connection. The pool is limited too: the system can only find donors who have already registered, and blood inventory is encoded manually by staff rather than linked to laboratory equipment or existing hospital and Red Cross information systems.

Because the transaction log is a hash-chained table in the system's own database with no distributed consensus, it can show that a record was altered but cannot stop someone with full database access from rebuilding the chain, so detection holds only while the chain is intact. It also cannot verify that an event actually happened outside the system. To respect donors' privacy rights, the log stores only references and hashes, not personal information, so records can still be corrected or deleted in the main database. Registering with the National Privacy Commission, which applies to organizations processing sensitive personal information of 1,000 or more individuals (National Privacy Commission, n.d.), and other formal data privacy steps belong to whichever organization eventually operates the system.

The rewards feature is limited to small-value vouchers and recognition. Small tokens and refreshments are compatible with voluntary, non-remunerated donations, but payment in cash or in kind that could substitute for money is not (World Health Organization, 2010). Partner sponsorship is voluntary, each voucher has a small maximum value, and points cannot be converted to cash or traded. Points are awarded only after staff confirm a completed donation, in keeping with the humanitarian principle of the National Blood Services Act (Republic Act No. 7719, 1994).

Finally, the system is evaluated through black box testing only. This shows that the functions behave as specified, but not code-level quality, performance under heavy load, or the real-world effect on donor turnout and response time. Testing uses simulated data, meaning generated donor profiles and requests, not real donor records, and involves 2 testers for each user role. The app is available in English only for now, and the prototype is tested in a single setting, Bulacan Medical Center, not deployed nationwide.

1.5 Significance of the Study

Recipients and their families. In an urgent request, the time spent searching for a donor is time the Recipient spends waiting. Dugo+ shortens that search by querying the donor pool automatically, so families no longer depend on calls and social media posts. It also eases the pressure on relatives to find blood on their own, a practice the Department of Health has flagged as risky (Philippine Information Agency, 2025). The World Health Organization (2010) notes that this pressure falls on families already under stress from the patient's illness.

Donors. Donors get one place to see their profile and donation history, and they are alerted only when a request fits their blood type, eligibility, and location. This matters because a study of Metro Manila adults found that respondents knew least about the interval between successive donations (Mappala et al., 2023). The same study identifies younger people as promising donors to recruit and retain. A pilot survey of U.S. adults aged 18 to 39 found that about 73% were willing to use an app to track their blood donations (Li et al., 2023), which suggests a readiness worth testing among Filipino donors. Repeat donors also receive recognition through the rewards feature.

Hospitals and blood banks. Staff work from a single searchable record, which addresses the slow, error-prone retrieval and reporting problem described by Eder (2018). The inventory feature and request-tracking dashboard show what is on hand and who has responded.

The Philippine Red Cross and the DOH. A searchable, eligibility-aware pool of willing donors supports the DOH goal of widening the donor base and the voluntary donation mandate of Republic Act No. 7719.

Administrators and oversight. Because requests, matches, donations, and point awards are logged in a tamper-evident record, administrators and partner institutions can trace what happened in any transaction, which spreadsheets and paper logs do not allow. The chained-hash approach borrows from blockchain-based designs proposed for the same traceability reasons (Sadri et al., 2021), in a form light enough for a single-hospital prototype.

Reach. About 98 million Filipinos were online by the end of 2025, and mobile connections were equal to 117 percent of the population (DataReportal, 2025). That makes a mobile-first solution realistic, and SMS alerts still reach donors whose connections do not include data.

Future researchers and developers. The largest share of the apps and studies reviewed by Li et al. (2023) focus on finding donors and blood centers (44.1%), while only 8.8% address retaining donors. Dugo+ gives later studies a documented, working baseline to extend, for example with a live pilot deployment.

CHAPTER 2

SYSTEMS ANALYSIS AND DESIGN

This chapter translates the documented problem into validated requirements, improved business processes, data and behavior models, and a technical design for the proposed system.

2.1 Requirements Analysis

2.1.1 Functional Requirements

Account Management (Donor & Recipient) Requirements

ID	Functional Requirement
FR-01	The system shall allow Donors and Recipients to register an account by providing personal details, blood type, and contact information, with email verification through a confirmation link.
FR-02	The system shall allow Donors and Recipients to log in using their registered credentials and maintain a secure session.
FR-03	The system shall allow Donors and Recipients to log out of their accounts, ending the active session securely.
FR-04	The system shall allow Donors and Recipients to create and edit a profile containing blood type, location, date of birth, and eligibility information.

Donor Requirements

ID	Functional Requirement
FR-005	The system shall allow Donors to view a list of available blood requests filtered by blood type compatibility and proximity, using the tiered search radius defined in NFR-05.
FR-006	The system shall allow Donors to accept or decline a blood request, updating the request status in real time.
FR-007	The system shall send emergency notifications to Donors email when a compatible and eligible match is found nearby.
FR-008	The system shall allow Donors to track their donation history, including dates, locations, and hospitals or blood banks involved.
FR-009	The system shall allow Donors to view accumulated points and rewards earned from completed donations.
FR-010	The system shall allow Donors to view a health tracker showing donation eligibility status and recommended next donation date.
FR-023	The system shall display the precise geolocation of all hospitals within donor radius on the map.
FR-024	The system shall prevent a donor from registering for a new Whole Blood Donation until 56 days (8 weeks) have elapsed since their last Whole Blood Donation.

FR-025	The system shall prevent a donor from registering for a new Platelet Donation until 7 days have elapsed since their last Platelet Donation.
FR-026	The system shall prevent a donor from registering for a new Plasma Donation until 28 days have elapsed since their last Plasma Donation.
FR-027	The system shall prevent a donor from registering for a new Double Red Cell Donation until 112 days (16 weeks) have elapsed since their last Double Red Cell Donation.
FR-028	The system shall display the relevant donation eligibility guidelines (e.g., recent surgery, transfusion, tattoo/piercing, pregnancy) to the donor based on their selected donation type before they can accept a blood request.
FR-029	The system shall require the donor to check an acknowledgment box confirming they have read and meet the displayed eligibility guidelines before they can accept a blood request.
FR-030	The system shall record the date and time of the donor's guideline acknowledgment as part of the donor's request history.
FR-036	The system shall automatically award reward points to a Donor's account once Hospital staff confirm a completed donation, per FR-033.
FR-038	The system shall obtain and record a Donor's consent, at registration or at match time, before their profile information is shared with or accessed by a Hospital, per NFR-16.

Recipient Requirements

ID	Functional Requirement
FR-011	The system shall allow Recipients to submit a blood request specifying blood type, quantity, urgency level, and hospital location.

Hospital Requirements

ID	Functional Requirement
FR-012	The system shall allow Hospital staff to log in and log out using role-based hospital credentials.
FR-013	The system shall allow Hospital staff to edit and maintain the hospital's profile information.
FR-014	The system shall allow Hospital staff to view the total number of blood units collected over a selected period.
FR-015	The system shall allow Hospital staff to view the hospital's location on a map for donor navigation purposes.
FR-016	The system shall allow Hospital staff to view a list of currently active blood requests submitted within their facility.
FR-017	The system shall allow Hospital staff to create blood requests on behalf of Recipients and route them to matching donors.
FR-018	The system shall allow Hospital staff to manage the hospital's subscription plan for continued system access.
FR-020	The system shall allow hospital staff to check current blood inventory levels and view the hospital's blood donation history.
FR-021	The system shall allow hospitals to submit emergency blood donation requests based on blood type compatibility and proximity, using the tiered search radius defined in NFR-05.

FR-031	The system shall route all matched donors to the hospital for final eligibility verification (physical screening, blood tests, and medical history check) prior to the actual donation.
FR-032	The system shall allow Hospital staff to mark a matched donor's status as "Cleared," "Deferred," or "Rejected" after conducting their own eligibility screening.
FR-033	The system shall update the donor's donation history only after Hospital staff confirms a completed donation, ensuring FR-024–FR-027 intervals are based on verified data, not self-reported claims.
FR-034	The system shall allow Hospital staff to add or update blood inventory records after a completed donation or issued transfusion.

System Administration Requirements

ID	Functional Requirement
FR-040	The system shall allow System Admin to create Hospital accounts and to suspend or remove Hospital and Donor/Recipient accounts as needed. Donor and Recipient accounts shall continue to be created via self-registration per FR-001.
FR-041	The system shall allow System Admin to configure system-wide parameters, including the tiered matching radius defined in NFR-05 and notification retry settings.
FR-046	The system shall allow System Admin to log in and log out using role-based administrator credentials.
FR-047	The system shall allow System Admin to view their own account profile.

System Wide/Cross Cutting Requirements

ID	Functional Requirement
FR-019	The system shall secure donor and recipient information by enforcing access controls and encryption in compliance with the Data Privacy Act of 2012 (RA 10173).
FR-022	The system shall display the precise geolocation of donors and recipients after having a match.
FR-035	The system shall allow authorized users (Admin/Hospital) to view the blockchain-based transaction/audit log of donation records and data access events, as maintained per NFR-15.
FR-037	The system shall prioritize higher-urgency blood requests, as specified in FR-011, FR-017, and FR-021, ahead of lower-urgency requests in the donor matching queue and the Hospital dashboard's request list.
FR-039	The system shall allow Donors and Recipients to submit a request to access, correct, or delete their personal data, and allow System Admin to process such requests, per NFR-19.
FR-042	The system shall commit a new blockchain record for every donation confirmation, inventory update, and data-access event, recorded at the time the action occurs.
FR-043	The system shall prevent modification or deletion of committed blockchain records; any correction shall be recorded as a new linked entry rather than an edit.
FR-044	The system shall allow System Admin to verify that a given audit record has not been altered by checking its hash against the chain.

FR-045	The system shall store only hashed or non-identifying references to personal data within blockchain audit records, ensuring that donor and recipient data deletion requests (FR-039) can be fulfilled without compromising blockchain record immutability (FR-043).
FR-048	The system shall allow Donors, Recipients, and Hospital staff to reset their password and recover account access through a verified channel (e.g. registered email). System Admin accounts are excluded, as they are premade and managed directly by the development team.

2.1.4 Non-Functional Requirements

Non-functional requirements describe how well the system must operate. Avoid vague statements such as “fast,” “secure,” or “user-friendly”; use measurable acceptance criteria whenever possible.

ID	Category	Non-Functional Requirement
NFR-01	Performance	The system shall match a submitted blood request with eligible nearby donors within 3 seconds of submission.
NFR-02	Performance	The system shall send an in app and email notifications to matched donors within 3 seconds of a match being found.
NFR-03	Performance	The system shall support at least 500 concurrent users without noticeable performance degradation.
NFR-04	Performance	The system shall load the donor dashboard, request list, and hospital dashboard within 3 seconds under normal network conditions.
NFR-05	Performance	The system's GPS-based matching engine shall process a search query matching donors to nearby blood requests and blood requests to nearby donors using a tiered search radius (starting at 5 km, expanding to 10 km, then 15 km, and capped at 20-25 km if no match is found), completing the search in under 5 seconds.
NFR-06	Reliability	The system shall be available 99.5% of the time to support blood requests occurring at any hour, including emergencies.
NFR-07	Reliability	The system shall retain a blood request as active and retry notification delivery if the initial alert and email fails.
NFR-08a	Reliability	The system shall perform automated backups of all donor, recipient, and hospital data to cloud storage to prevent data loss in the event of a server failure.
NFR-08b	Reliability	The system shall perform full data backups at least once every 24 hours, with active blood requests and match transactions backed up in real time.
NFR-09	Reliability	The system shall gracefully handle third-party service outages (email, GPS) and inform the user when a dependent service is unavailable.
NFR-10	Security	The system shall encrypt all sensitive personal and health-related data both at rest and in transit.
NFR-11	Security	The system shall restrict access to donor information to authorized hospital users and System Admin only, verified through role-based access control.
NFR-12	Security	The system shall require email verification for account registration to prevent fake or duplicate accounts.

NFR-13	Security	The system shall enforce a minimum password length and complexity requirement of at least 8 characters.
NFR-14 (DEFFERED)	Security	The system shall automatically end a user session after a defined period of inactivity.
NFR-15	Security	The system shall maintain a tamper-evident blockchain-based audit trail of donation records, inventory updates, and donor information access.
NFR-16	Security	The system shall obtain and record donor consent before donor information is shared with or accessed by a hospital.
NFR-17	Compliance	The system shall comply with the Data Privacy Act of 2012 (Republic Act No. 10173) in the collection, storage, and processing of personal data.
NFR-18	Compliance	The system shall comply with DOH Administrative Order No. 2020-0030 on data privacy guidelines for health information.
NFR-19	Compliance	The system shall provide a mechanism for users to request access to, correction of, or deletion of their personal data.
NFR-20	Usability	The system's mobile web app interface shall be simple and intuitive for first-time users with minimal technical literacy.
NFR-21	Usability	The system shall display clear, immediate error messages in plain language.
NFR-22	Usability	The system shall support use in low-connectivity conditions, with clear messaging when internet-dependent features are unavailable.
NFR-23	Usability	The Hospital dashboard shall present request status, inventory, and donor matches in a clear, scannable layout for time-pressured staff.
NFR-24	Scalability	The system's architecture shall support future expansion to additional hospitals and geographic areas without redesigning the core matching engine.
NFR-25	Scalability	The centralized donor database shall accommodate a growing number of donor profiles and records without significant performance loss.
NFR-26	Maintainability	The system shall use a modular architecture allowing individual features to be updated independently.
NFR-27	Maintainability	The system shall log system errors and exceptions to support debugging and maintenance.
NFR-28	Compatibility	Donor and Recipient access shall be via Android/iOS mobile web app.
NFR-29	Compatibility	Hospital and System Admin access shall be via a web browser-based dashboard.

2.2 Business Process Analysis

Student guidance: *Examine how work is currently performed before deciding what to automate. Document actors, steps, inputs, outputs, rules, delays, errors, duplication, and control points. Present both the existing AS IS process and the improved TO BE process.*

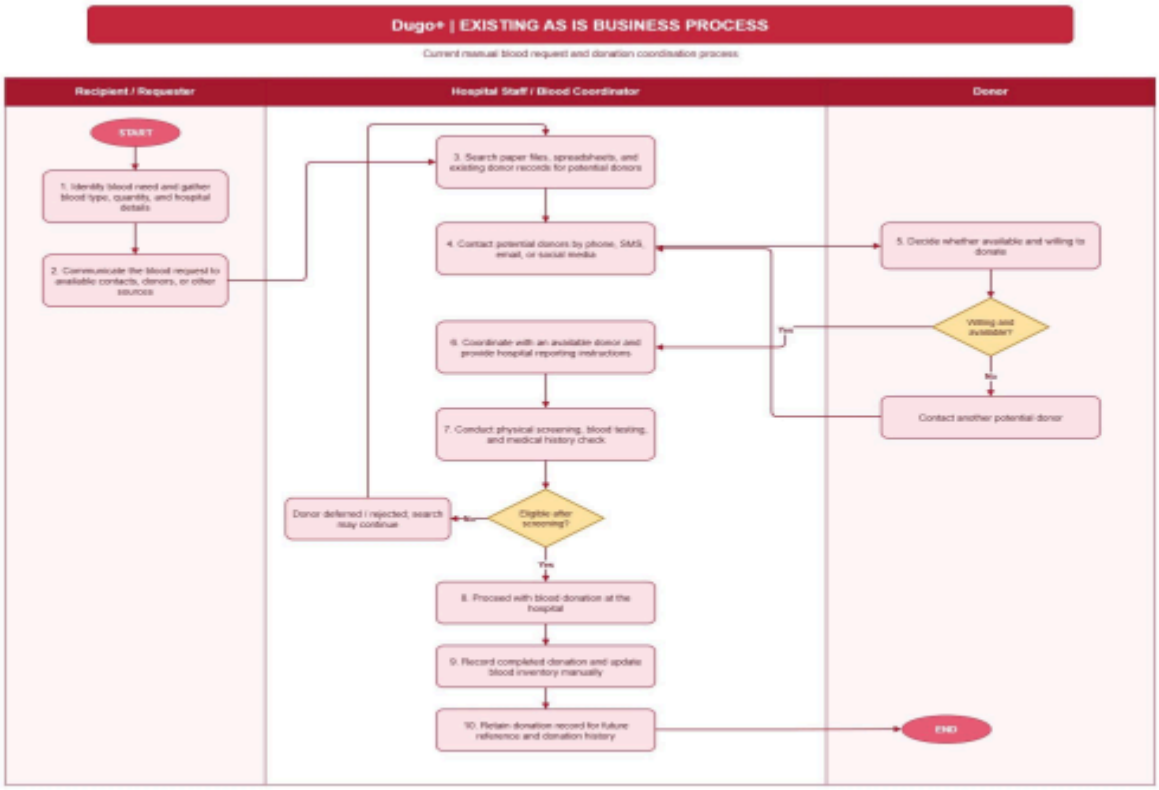
2.2.1 Existing AS IS Process

[Describe the current process from start to finish. Explain who performs each activity, what information or document is used, and how the process ends.]

Step	Responsible Actor	Input	Activity	Output	Problem or Bottleneck
1	Recipient / Hospital Staff	Blood requirement	Recipient or hospital staff identifies the need for blood and gathers the required blood type, quantity, and hospital information.	Blood request information	Request information may be incomplete or require manual coordination.
2	Recipient / Hospital Staff	Blood request information	The blood request is communicated to available contacts, donors, or other sources.	Blood request communicated	Communication may be slow, especially during urgent situations.
3	Hospital Staff / Blood Coordinator	Required blood type and donor records	Hospital staff or blood coordinators search paper files, spreadsheets, or existing donor records for potential donors with the required blood type.	List of potential donors	Donor information is scattered across multiple records, making compatible donors difficult and time-consuming to identify.
4	Hospital Staff / Blood Coordinator	Donor contact information	Potential donors are contacted through phone calls,email, or social media and asked about their availability to donate.	Donor response	Contacting multiple donors through separate communication channels can cause delays.
5	Donor	Blood request	The donor decides whether they are available and willing to respond to the blood request.	Accepted or declined response	Donor responses may not be received immediately.

6	Hospital Staff	Donor information	Hospital staff coordinate with an available donor and provide instructions for reporting to the hospital.	Donor scheduled for screening	Coordination depends on manual communication.
7	Hospital Staff	Donor personal and medical information	Hospital staff conduct physical screening, blood testing, and medical history checking to determine donor eligibility.	Cleared, deferred, or rejected donor	Eligibility verification occurs only after the donor reaches the hospital.
8	Donor / Hospital Staff	Cleared donor	The donor proceeds with the blood donation at the hospital.	Completed donation	Donation information requires manual recording and updating.
9	Hospital Staff	Donation information	Hospital staff record the completed donation and update the hospital's blood inventory.	Updated donation and inventory records	Manual record updates can cause delays or inaccurate records.
10	Hospital Staff	Donation record	Hospital staff retain the donation information for future reference, including the donor's previous donation activity.	Donation history	Maintaining accurate historical records is difficult when records are manually managed.

Figure 2.1. Existing AS IS Business Process



2.3 Data Flow Diagram and Context Diagram

Student guidance: *Show how data moves among external entities, processes, and data stores. Start with a context diagram containing one process for the entire system, then decompose it into Level 0 and lower-level DFDs as needed. Balance inputs and outputs between levels. Do not show program control flow.*

Symbol or Element	Project-Specific Meaning
External entity	Donor, Recipient, and Hospital staff
Process	Manage Account, Submit Request, Match Donor, Manage Inventory, Send Notification
Data flow	Profile, Request Acceptance, Emergency Blood Request Notification, Blood Request, Status Update, Request, Matches, Donor Data, Request data, Match found , Clearance Status, Hospital Request, Request Info, Donor Profile, Compatible donors, Nearby donors, Eligible donors, Match Record
Data store	**D1 Donor DB** , **D2 Request DB** , **D3 MATCH DB**

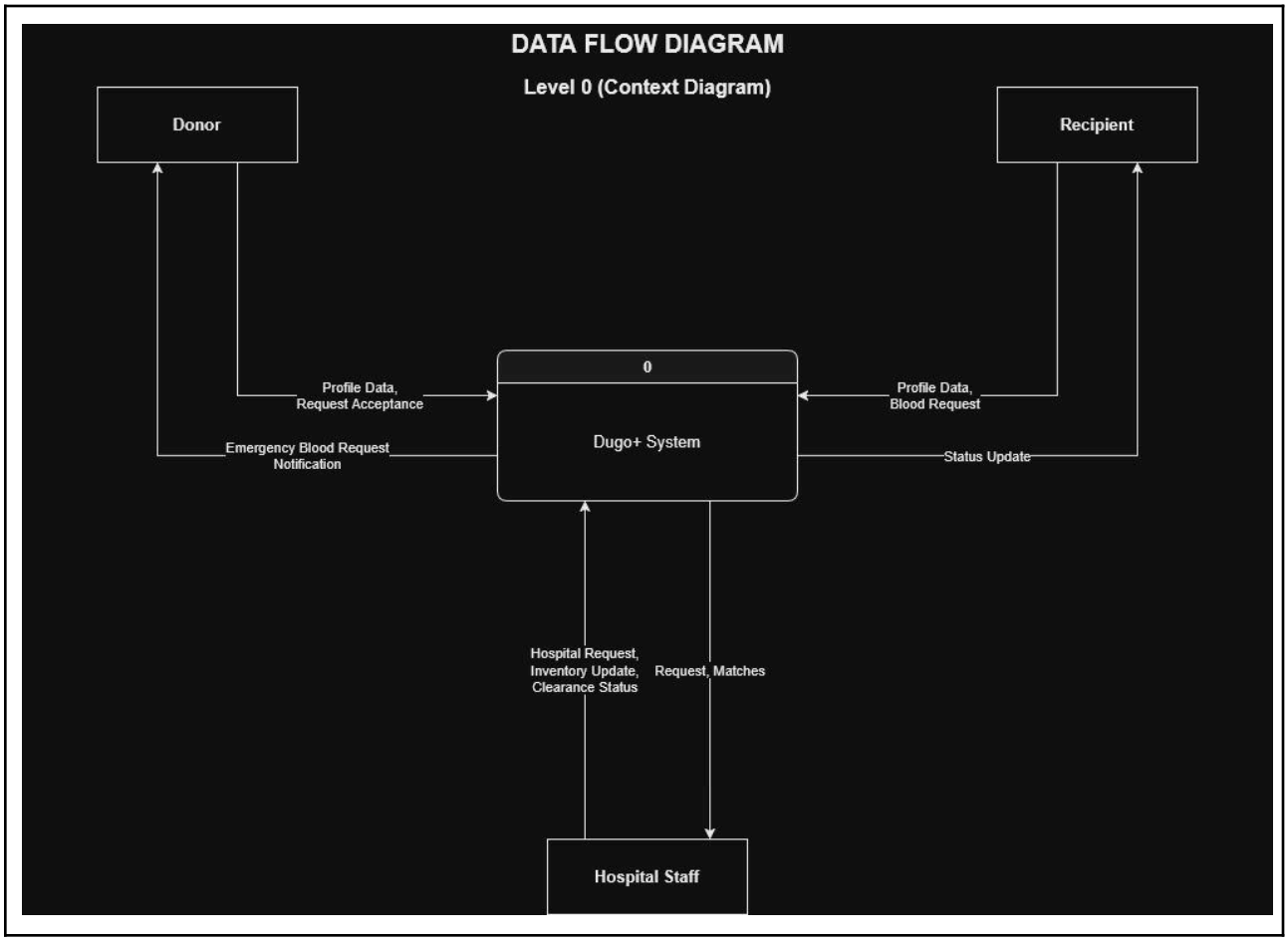


Figure 2.3. Context Diagram of the Proposed System

The context diagram treats Dugo+ as a single system boundary that handles all internal functions, including account management, request handling, donor matching, and notifications. Three external entities interact with the system, Donor, Recipient, and Hospital Staff. Donors send profile data and receive match alerts, while Recipients submit blood requests and receive status updates. Hospital Staff send inventory updates and donor clearance confirmations, and in return receive matches and notifications. All detailed processing, such as matching logic and eligibility checks, is handled internally and further broken down in the Level 1 diagram.

Lower-Level DFDs

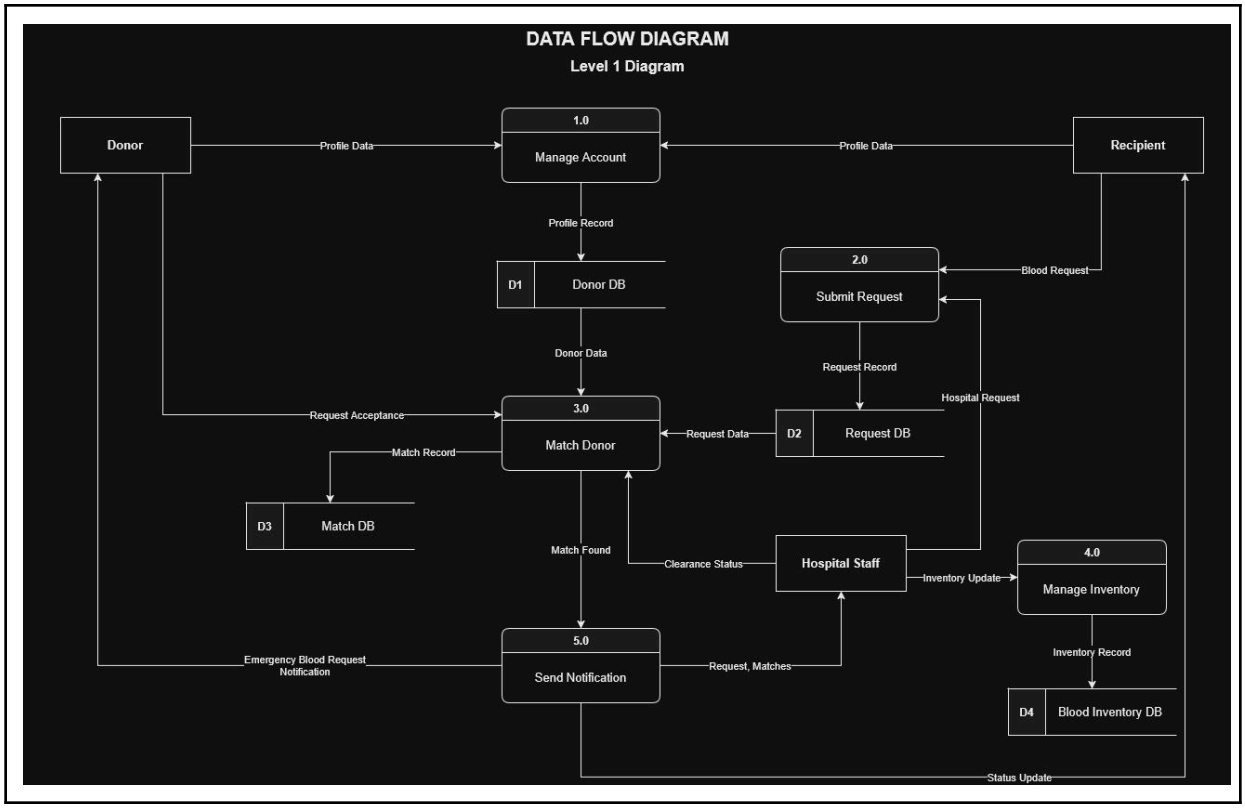


Figure 2.5. Level 1 Data Flow Diagram for Process 1.0 Manage Account, Process 2.0 Submit Request, 3.0 Match Donor, 4.0 Manage Inventory and 5.0 Send Notification

Lower-Level DFDs

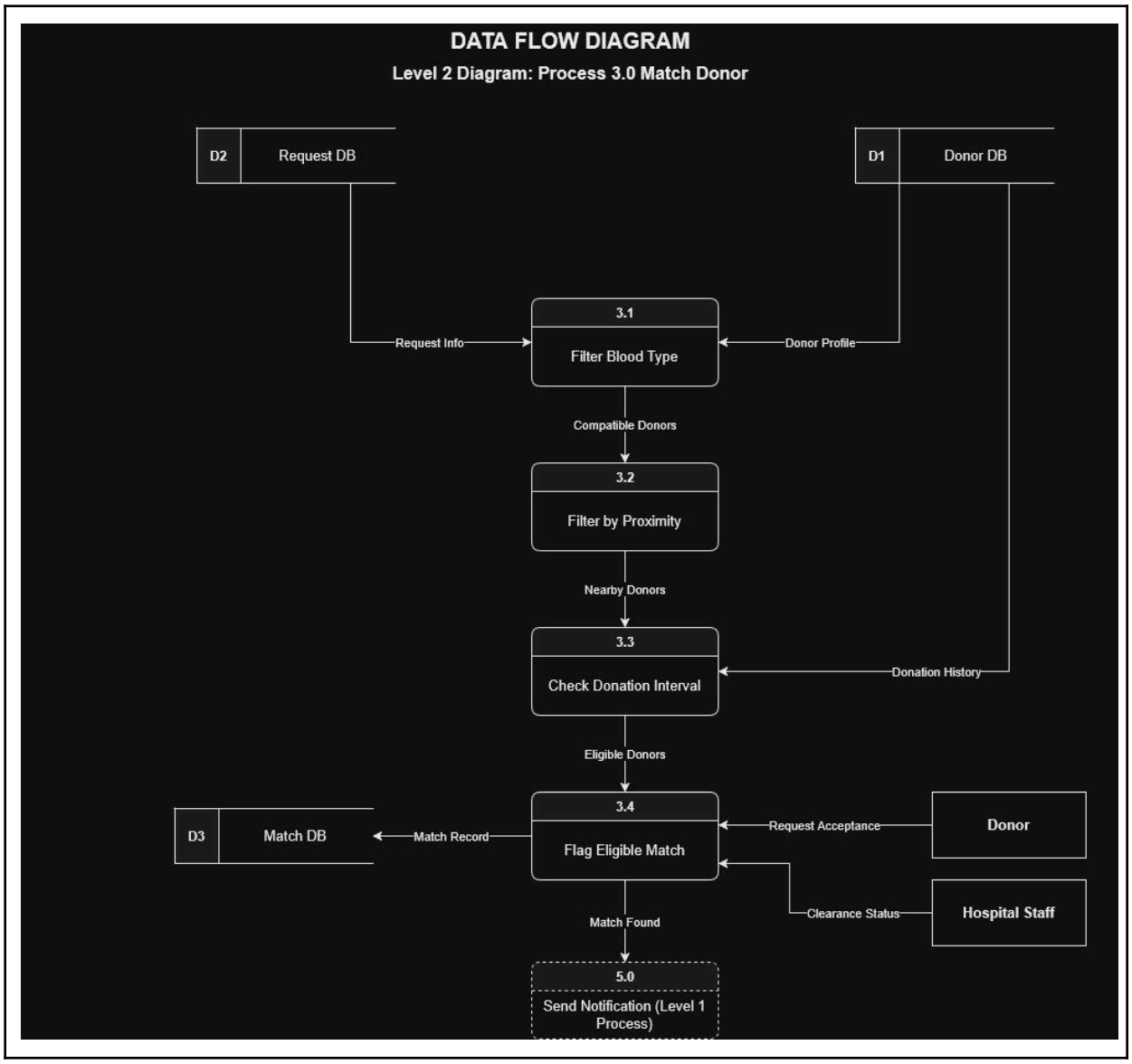


Figure 2.5. Level 2 Data Flow Diagram for Process 3.0 Match Donor

2.4 Entity Relationship Diagram

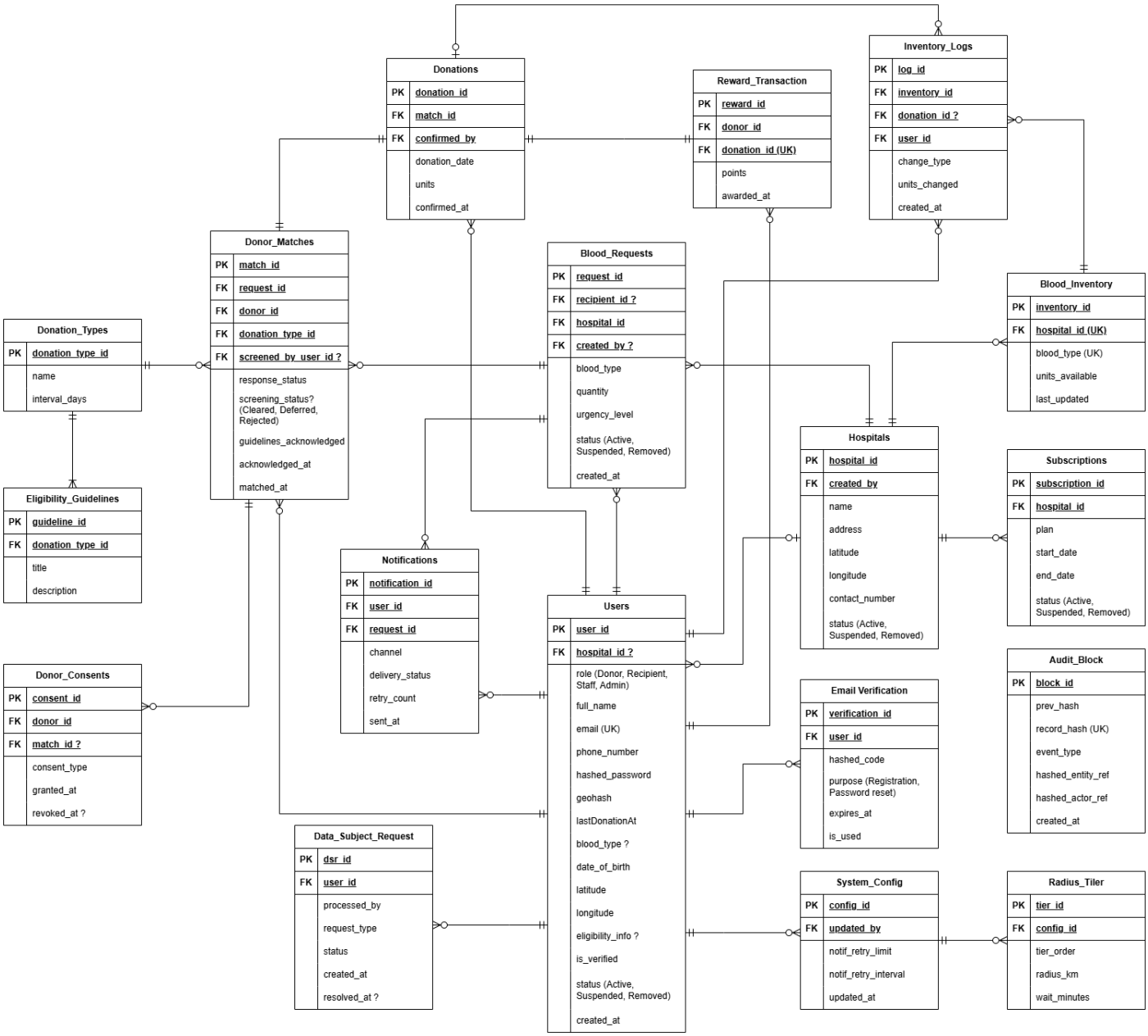


Figure 2.6. Entity Relationship Diagram of the Proposed System

2.4.1 Entity Description

Entity	Purpose	Primary Key	Important Attributes	Related Entities
Users	Stores all system accounts (donors, recipients, staff, admins)	user_id	role, full_name, email, phone_number, hashed_password,geohash lastDonationAt, blood_type, date_of_birth, latitude, longitude, eligibility_info, is_verified, status, hospital_id	Hospitals, Blood_Requests, Donor_Matches, Donations, Reward_Transaction, Inventory_Logs, Notifications,

				Data_Subject_Request, Donor_Consents, System_Config
Hospitals	Stores partner hospital details and locations	hospital_id	name, address, latitude, longitude, contact_number, status, created_by	Users, Subscriptions, Blood_Requests, Blood_Inventory
Subscriptions	Tracks a hospital's subscription plan and validity	subscription_id	hospital_id, plan, start_date, end_date, status	Hospitals
Blood_Requests	Records emergency or regular blood requests posted by or for hospitals	request_id	recipient_id, hospital_id, created_by, blood_type, quantity, urgency_level, status, created_at	Hospitals, Users, Donor_Matches, Notifications
Donor_Matches	Links a donor to a blood request and tracks their response and screening	match_id	request_id, donor_id, donation_type_id, screened_by_user_id, response_status, screening_status, guidelines_acknowledged, acknowledged_at, matched_at	Blood_Requests, Users, Donation_Types, Donations, Donor_Consents
Donation_Types	Defines kinds of donations and their waiting intervals	donation_type_id	name, interval_days	Donor_Matches, Eligibility_Guidelines
Eligibility_Guidelines	Holds eligibility rules shown to donors per donation type	guideline_id	donation_type_id, title, description	Donation_Types
Donor_Consents	Records consents granted or revoked by donors (data privacy)	consent_id	donor_id, match_id, consent_type, granted_at, revoked_at	Users, Donor_Matches
Donations	Records completed and confirmed donations	donation_id	match_id, confirmed_by, donation_date, units, confirmed_at	Donor_Matches, Users, Reward_Transaction, Inventory_Logs
Reward_Transaction	Tracks points awarded to donors for donations	reward_id	donor_id, donation_id, points, awarded_at	Users, Donations
Blood_Inventory	Tracks blood stock per blood type for each hospital	inventory_id	hospital_id (UK), blood_type (UK), units_available, last_updated	Hospitals, Inventory_Logs
Inventory_Logs	Keeps a history of every stock change	log_id	inventory_id, donation_id, user_id, change_type,	Blood_Inventory, Donations,

			units_changed, created_at	Users
Notifications	Logs alerts sent to users about blood requests	notification_id	user_id, request_id, channel, delivery_status, retry_count, sent_at	Users, Blood_Requests
Data_Subject_Request	Tracks user data privacy requests (access, deletion, etc.)	dsr_id	user_id, processed_by, request_type, status, created_at, resolved_at	Users
System_Config	Holds global settings such as notification retry rules	config_id	updated_by, notif_retry_limit, notif_retry_interval, updated_at	Users, Radius_Tiler
Radius_Tiler	Defines search radius tiers and wait times for donor matching	tier_id	config_id, tier_order, radius_km, wait_minutes	System_Config
Audit_Block	Tamper-evident, hash-chained audit trail of system events	block_id	prev_hash, record_hash (UK), event_type, hashed_entity_ref, hashed_actor_ref, created_at	None (standalone)

2.4.2 Data Dictionary

Table or Entity	Field	Data Type and Size	Key	Null	Description or Validation
Users	user_id	INT	PK	No	Auto-generated unique user ID
Users	hospital_id	INT	FK	Yes	References Hospitals. Set only for Staff/Admin tied to a hospital
Users	role	ENUM('Donor','Recipient','Staff','Admin')		No	Account role, must be one of the listed values
Users	full_name	VARCHAR(100)		No	User's full name, letters and spaces only
Users	email	VARCHAR(255)	UK	No	Must be a valid email format and unique
Users	phone_number	VARCHAR(20)		No	Valid PH mobile number format (e.g., 09XXXXXXXXX or +639XXXXXXXXX)
Users	hashed_pas	VARCHAR(255)		No	Securely hashed password,

	sword				never stored as plain text
Users	geohash				
Users	lastDonationAt				
Users	blood_type	ENUM('A+','A-','B+','B-','AB+','AB-','O+','O-')		Yes	Required for donors and recipients
Users	date_of_birth	DATE		No	Must not be a future date. Donors must meet the minimum donation age
Users	latitude	DECIMAL(9,6)		No	Range -90 to 90
Users	longitude	DECIMAL(9,6)		No	Range -180 to 180
Users	eligibility_info	TEXT		Yes	Donor's self-declared eligibility from date of birth and last donation info
Users	is_verified	BOOLEAN		No	Default FALSE. Set TRUE after email verification
Users	status	ENUM('Active','Suspended','Removed')		No	Default 'Active'
Users	created_at	TIMESTAMP		No	Default current timestamp
Hospitals	hospital_id	INT	PK	No	Auto-generated unique hospital ID
Hospitals	created_by	INT	FK	No	References Users (the admin who registered the hospital)
Hospitals	name	VARCHAR(150)		No	Hospital name
Hospitals	address	VARCHAR(255)		No	Complete street address
Hospitals	latitude	DECIMAL(9,6)		No	Range -90 to 90
Hospitals	longitude	DECIMAL(9,6)		No	Range -180 to 180
Hospitals	contact_number	VARCHAR(20)		No	Valid landline or mobile number format
Hospitals	status	ENUM('Active','Suspended','Removed')		No	Default 'Active'
Subscriptions	subscription_id	INT	PK	No	Auto-generated unique subscription ID
Subscriptions	hospital_id	INT	FK	No	References Hospitals

Subscriptions	plan	VARCHAR(50)		No	Subscription plan name
Subscriptions	start_date	DATE		No	Start of the subscription
Subscriptions	end_date	DATE		No	Must be later than start_date
Subscriptions	status	ENUM('Active','Suspended','Removed')		No	Default 'Active'
Blood_Requests	request_id	INT	PK	No	Auto-generated unique request ID
Blood_Requests	recipient_id	INT	FK	Yes	References Users (the recipient, if registered)
Blood_Requests	hospital_id	INT	FK	No	References Hospitals
Blood_Requests	created_by	INT	FK	Yes	References Users (staff who created the request)
Blood_Requests	blood_type	ENUM('A+','A-','B+','B-','AB+','AB-','O+','O-')		No	Blood type needed
Blood_Requests	quantity	INT		No	Units needed, must be greater than 0
Blood_Requests	urgency_level	VARCHAR(20)		No	Urgency label (e.g., Low, Medium, High, Critical)
Blood_Requests	status	ENUM('Active', 'Matched', 'Completed', 'Cancelled, 'Expired')		No	Default 'Active'
Blood_Requests	created_at	TIMESTAMP		No	Default current timestamp
Donor_Matches	match_id	INT	PK	No	Auto-generated unique match ID
Donor_Matches	request_id	INT	FK	No	References Blood_Requests
Donor_Matches	donor_id	INT	FK	No	References Users (role must be Donor)
Donor_Matches	donation_type_id	INT	FK	No	References Donation_Types
Donor_Matches	screened_by_user_id	INT	FK	Yes	References Users (staff who screened the donor)
Donor_Matc	response_st	VARCHAR(20)		No	Donor's response (e.g.,

hes	atus				Pending, Accepted, Declined)
Donor_Matches	screening_status	ENUM('Cleared','Deferred','Rejected')		Yes	Set after screening by staff
Donor_Matches	guidelines_acknowledged	BOOLEAN		No	Default FALSE. Must be TRUE before screening
Donor_Matches	acknowledged_at	TIMESTAMP		Yes	Required when guidelines_acknowledged is TRUE
Donor_Matches	matched_at	TIMESTAMP		No	Default current timestamp
Donation_Types	donation_type_id	INT	PK	No	Auto-generated unique type ID
Donation_Types	name	VARCHAR(50)		No	Type name (e.g., Whole Blood, Platelets)
Donation_Types	interval_days	INT		No	Minimum days between donations, must be greater than 0
Eligibility_Guidelines	guideline_id	INT	PK	No	Auto-generated unique guideline ID
Eligibility_Guidelines	donation_type_id	INT	FK	No	References Donation_Types
Eligibility_Guidelines	title	VARCHAR(100)		No	Short guideline title
Eligibility_Guidelines	description	TEXT		No	Full guideline text shown to donors
Donor_Consents	consent_id	INT	PK	No	Auto-generated unique consent ID
Donor_Consents	donor_id	INT	FK	No	References Users
Donor_Consents	match_id	INT	FK	Yes	References Donor_Matches. Null for general consents
Donor_Consents	consent_type	VARCHAR(50)		No	Type of consent given (e.g., Data Sharing, Screening)
Donor_Consents	granted_at	TIMESTAMP		No	When consent was granted
Donor_Consents	revoked_at	TIMESTAMP		Yes	Null while consent is active. Must be after granted_at
Donations	donation_id	INT	PK	No	Auto-generated unique

					donation ID
Donations	match_id	INT	FK, UK	No	References Donor_Matches. One donation per match
Donations	confirmed_by	INT	FK	No	References Users (staff who confirmed)
Donations	donation_date	DATE		No	Must not be a future date
Donations	units	INT		No	Units donated, must be greater than 0
Donations	confirmed_at	TIMESTAMP		No	When staff confirmed the donation
Reward_Transaction	reward_id	INT	PK	No	Auto-generated unique reward ID
Reward_Transaction	donor_id	INT	FK	No	References Users
Reward_Transaction	donation_id	INT	FK, UK	No	References Donations. One reward per donation
Reward_Transaction	points	INT		No	Points awarded, must be greater than 0
Reward_Transaction	awarded_at	TIMESTAMP		No	Default current timestamp
Blood_Inventory	inventory_id	INT	PK	No	Auto-generated unique inventory ID
Blood_Inventory	hospital_id	INT	FK, UK*	No	References Hospitals
Blood_Inventory	blood_type	ENUM('A+','A-','B+','B-','AB+','AB-','O+','O-')	UK*	No	*Unique together with hospital_id
Blood_Inventory	units_available	INT		No	Current stock, cannot be negative
Blood_Inventory	last_updated	TIMESTAMP		No	Auto-updated on every stock change
Inventory_Logs	log_id	INT	PK	No	Auto-generated unique log ID
Inventory_Logs	inventory_id	INT	FK	No	References Blood_Inventory
Inventory_Logs	donation_id	INT	FK	Yes	References Donations. Set only for donation-based additions
Inventory_Lo	user_id	INT	FK	No	References Users (who

gs					made the change)
Inventory_Logs	change_type	VARCHAR(30)		No	Type of change (e.g., Donation, Usage, Adjustment, Expired)
Inventory_Logs	units_changed	INT		No	Positive for additions, negative for deductions
Inventory_Logs	created_at	TIMESTAMP		No	Default current timestamp
Notifications	notification_id	INT	PK	No	Auto-generated unique notification ID
Notifications	user_id	INT	FK	No	References Users (recipient of the alert)
Notifications	request_id	INT	FK	No	References Blood_Requests
Notifications	channel	VARCHAR(20)		No	Delivery channel (e.g., Push, Email)
Notifications	delivery_status	VARCHAR(20)		No	Status (e.g., Pending, Sent, Failed)
Notifications	retry_count	INT		No	Default 0. Must not exceed notif_retry_limit
Notifications	sent_at	TIMESTAMP		Yes	Null until successfully sent
Data_Subject_Request	dsr_id	INT	PK	No	Auto-generated unique request ID
Data_Subject_Request	user_id	INT	FK	No	References Users (the data subject)
Data_Subject_Request	processed_by	INT	FK	Yes	References Users (admin who handled it)
Data_Subject_Request	request_type	VARCHAR(30)		No	Type (e.g., Access, Correction, Deletion)
Data_Subject_Request	status	VARCHAR(20)		No	Status (e.g., Pending, In Progress, Resolved, Rejected)
Data_Subject_Request	created_at	TIMESTAMP		No	Default current timestamp
Data_Subject_Request	resolved_at	TIMESTAMP		Yes	Null until resolved
System_Config	config_id	INT	PK	No	Auto-generated unique config ID
System_Config	updated_by	INT	FK	No	References Users (admin who last changed it)

System_Config	notif_retry_limit	INT		No	Max notification retries, must be 0 or greater
System_Config	notif_retry_interval	INT		No	Minutes between retries, must be greater than 0
System_Config	updated_at	TIMESTAMP		No	Auto-updated on every change
Radius_Tiler	tier_id	INT	PK	No	Auto-generated unique tier ID
Radius_Tiler	config_id	INT	FK	No	References System_Config
Radius_Tiler	tier_order	INT		No	Search order (1 is searched first)
Radius_Tiler	radius_km	DECIMAL(5,2)		No	Search radius in kilometers, must be greater than 0
Radius_Tiler	wait_minutes	INT		No	Minutes to wait before expanding to the next tier
Audit_Block	block_id	BIGINT	PK	No	Auto-generated unique block ID
Audit_Block	prev_hash	CHAR(64)		Yes	Hash of the previous block. Null for the first block
Audit_Block	record_hash	CHAR(64)	UK	No	SHA-256 hash of this record, must be unique
Audit_Block	event_type	VARCHAR(50)		No	Type of event logged
Audit_Block	hashed_entity_ref	CHAR(64)		No	Hashed reference to the affected entity
Audit_Block	hashed_actor_ref	CHAR(64)		No	Hashed reference to the user who acted
Audit_Block	created_at	TIMESTAMP		No	Default current timestamp

2.5 Use Case Diagram

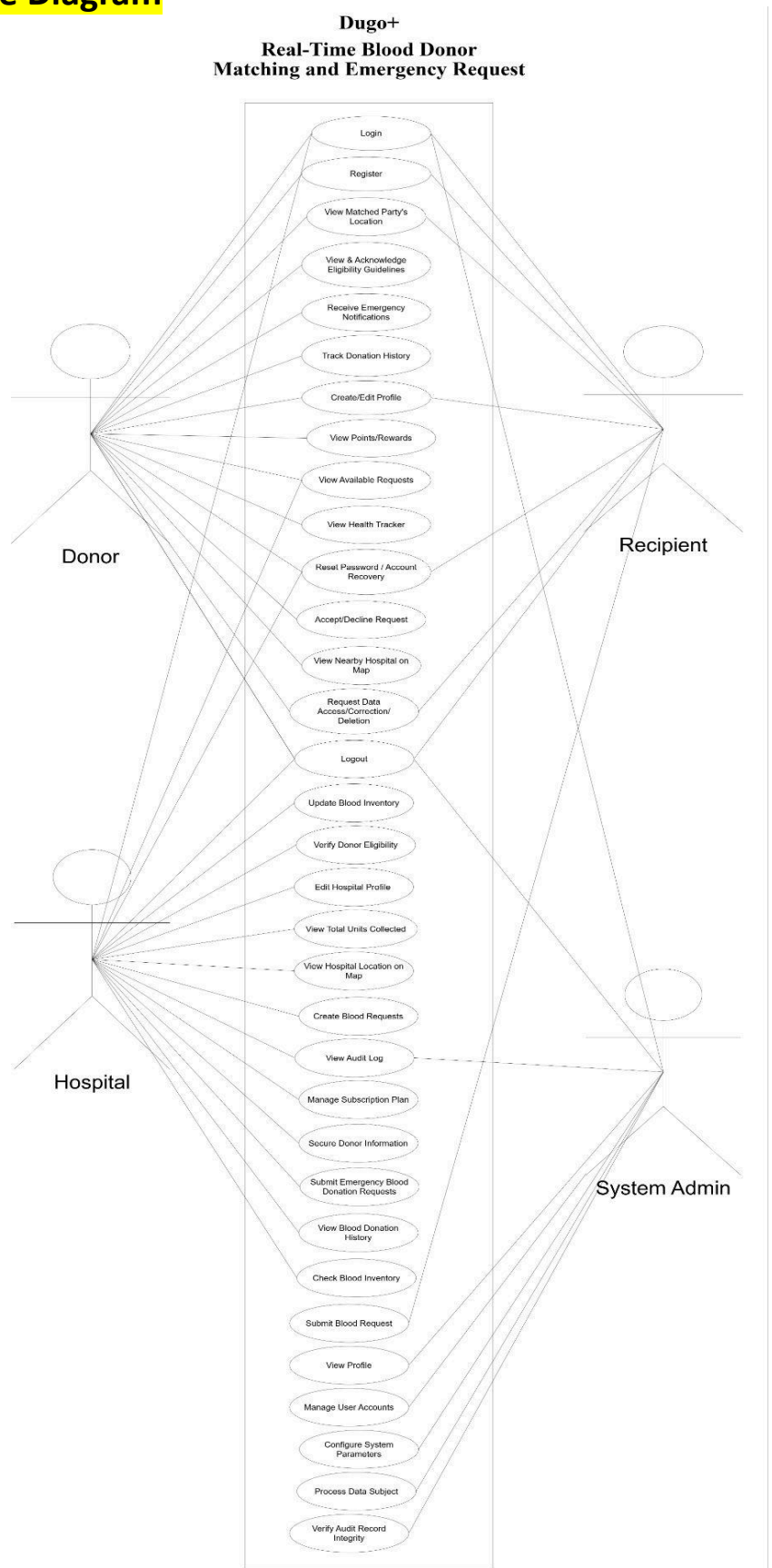


Figure 2.7. Use Case Diagram of the Proposed System

2.5.1 Use Case Summary

Use Case ID	Use Case Name	Primary Actor	Brief Description
UC-001	Register	Donor, Recipient	Allows a new Donor or Recipient to create a verified Dugo+ account using personal information, consent, and email verification.
UC-002	Login	Donor, Recipient, Hospital, System Admin	Allows users to securely access the functions and dashboard permitted for their account role.
UC-003	Create/Edit Profile	Donor, Recipient	Allows users to create or update personal information, including blood type, contact information, and location.
UC-004	View Available Requests	Donor	Allows Donors to view active blood requests filtered according to blood compatibility and proximity.
UC-005	View Health Tracker	Donor	Allows Donors to view donation eligibility information, waiting periods, and recommended next donation dates.
UC-006	Accept/Decline Request	Donor	Allows Donors to accept or decline an active blood request after reviewing the applicable eligibility guidelines.
UC-007	Submit Blood Request	Recipient	Allows Recipients to submit blood requests and initiate matching with compatible and nearby Donors.
UC-008	Track Donation History	Donor	Allows Donors to view their verified previous blood donation records
UC-009	Receive Emergency Notifications	Donor	Sends emergency alerts to compatible and eligible Donors through configured email channels.
UC-010	View Points/Rewards	Donor	Allows Donors to view their earned reward points and available rewards from completed donations.
UC-011	View Total Units Collected	Hospital	Allows Hospital staff to view the total number of blood units collected over a selected period.
UC-012	Create Blood Requests	Hospital	Allows Hospital staff to create blood requests on behalf of patients and initiate Donor matching.
UC-013	View Active Requests	Hospital	Allows Hospital staff to monitor active blood requests and their current status.
UC-014	View Blood	Hospital	Allows authorized Hospital staff to

	Donation History		review a Donor's recorded donation history
UC-015	Manage Subscription Plan	Hospital	Allows Hospital staff to view and manage the hospital's Dugo+ subscription plan.
UC-016	Secure Donor Information	Hospital	Allows Hospital staff to access Donor information only when authorized and applicable consent is recorded.
UC-017	Check Blood Inventory	Hospital	Allows Hospital staff to view current blood inventory by blood type, quantity, and other recorded details.
UC-018	Edit Hospital Profile	Hospital	Allows authorized Hospital staff to update hospital information such as facility details, contact information, and location.
UC-019	View Nearby Hospital on Map	Donor	Allows Donors to locate nearby hospitals using the system map and available location information.
UC-020	Submit Emergency Blood Donation Requests	Hospital	Allows Hospital staff to submit urgent blood requests, initiate matching, and trigger emergency notifications.
UC-021	Logout	Donor, Recipient, Hospital, System Admin	Allows authenticated users to securely end their current Dugo+ session.
UC-022	View Matched Party's Location	Donor, Recipient	Allows a Donor or Recipient to view the permitted location of the matched party after a successful match
UC-023	View and Acknowledge Eligibility Guidelines	Donor	Allows Donors to review applicable donation guidelines and record acknowledgment before accepting a request.
UC-024	Reset Password / Account Recovery	Donor, Recipient, Hospital	Allows users to recover account access by verifying their identity and resetting their password.
UC-025	Request Data Access/Correction/ Deletion	Donor, Recipient	Allows users to submit requests to access, correct, or delete their personal data held by Dugo+.
UC-026	Update Blood Inventory	Hospital	Allows Hospital staff to add or update inventory records following donations or transfusions and record the related audit event.
UC-027	Verify Donor Eligibility	Hospital	Allows Hospital staff to record the final Donor screening result as Cleared, Deferred, or Rejected.
UC-028	Configure System Parameters	System Admin	Allows the System Admin to configure system-wide matching radius and notification retry parameters.

UC-029	Manage User Accounts	System Admin	Allows the System Admin to create Hospital accounts and suspend or remove authorized user accounts
UC-030	Process Data Subject Request	System Admin	Allows the System Admin to review and process requests for personal data access,
UC-031	View Audit Log	Hospital, System Admin	Allows authorized users to review blockchain-based audit records for donation, inventory, and data-access events.
UC-032	Verify Audit Record Integrity	System Admin	Allows the System Admin to verify whether an audit record has been altered by checking its stored hash against the blockchain.
UC-033	View Profile	System Admin	Allows the System Admin to view their own administrator profile information.
UC-034	View Hospital Location on Map	Hospital	Allows Hospital staff to view the hospital's saved location on the system map.

2.5.2 Detailed Use Case Description

Field	Description
Use Case ID and Name	UC-001 — Register
Primary Actor	Donor, Recipient
Goal	Create an account to access Dugo+.
Trigger	The user selects Register from the login page.
Preconditions	The user has a mobile device, an internet connection, and a reachable contact number.
Main Flow	1. The user opens the registration form. 2. The user enters the required personal details, blood type, contact information, username, and password. The user then selects a role: Donor or Recipient. 3. The system presents a consent form for the user to read and agree to before their information is shared with the hospital. 4. The system validates the submitted information. 5. The system sends an email containing a verification link to the user's registered email address. 6. The user opens the email and clicks the verification link. 7. The system verifies the user's email address and creates the account with the selected role (Donor or Recipient). 8. The system confirms successful registration and directs the user to the login page.
Alternative or Exception Flow	3a. A required field is missing or a value is invalid: The system identifies the field and asks the user to correct it. 3b. The username is already in use: The system displays an error and requests a

	different username. 3c. The password does not meet the minimum length or complexity requirement: The system displays the password requirements. 4a. Email delivery fails: The system informs the user and offers to resend the verification email when available. 5a. The email verification link is invalid or expired: The system rejects the link and asks the user to request a new verification email.
Postconditions	Success: A verified Donor or Recipient account is created and is available for login. Failure: No account is created; entered information remains unregistered.

Field	Description
Use Case ID and Name	UC-002 — Login
Primary Actor	Donor, Recipient, Hospital, System Admin
Goal	Access the functions permitted for the user's account role.
Trigger	The user opens Dugo+ and the login page is displayed.
Preconditions	The user has an existing account and valid login credentials. Hospital and System Admin users use their assigned role-based credentials.
Main Flow	1. The user opens the login page. 2. The user enters the registered username and password or the assigned credentials. 3. The system validates the credentials and account status. 4. The system establishes an authenticated session. 5. The system displays the dashboard appropriate to the user's role.
Alternative or Exception Flow	2a. A required field is empty: The system prompts the user to complete it. 3a. The credentials are incorrect or the account cannot be used: The system displays an error without opening a session. 3b. The account is suspended or removed: The system denies access and displays an appropriate message. 3c. The internet connection is unavailable: The system reports the connection problem and does not establish a session.
Postconditions	Success: An authenticated session is active and the role-appropriate dashboard is displayed. Failure: No authenticated session is created.

Field	Description
Use Case ID and Name	UC-003 — Create/Edit Profile
Primary Actor	Donor, Recipient
Goal	Create or update personal profile information used by Dugo+.
Trigger	The user opens the profile page after logging in.

Preconditions	The user is logged in.
Main Flow	1. The user opens the profile page. 2. The system displays the existing profile, if one exists. 3. The user enters or edits the applicable personal details, including name, blood type, date of birth, contact information, location, and eligibility information. 4. The system validates the entered information. 5. The system saves the valid profile information. 6. The system displays the updated profile and a confirmation message.
Alternative or Exception Flow	3a. The user cancels the operation: The system discards unsaved changes. 3b. Location permission is unavailable: The system allows the user to provide a location manually, where supported. 4a. A required field is missing or a value is invalid: The system identifies the issue and requests correction. 5a. The information cannot be saved: The system displays an error and retains the previously saved profile.
Postconditions	Success: The user's profile is created or updated. Failure or cancellation: The previously saved profile remains unchanged.

Field	Description
Use Case ID and Name	UC-004 — View Available Requests
Primary Actor	Do nor
Goal	Browse active blood requests that may match the donor's blood type and proximity.
Trigger	The Donor opens the Available Requests section.
Preconditions	The Donor is logged in and has a profile containing blood type and location information.
Main Flow	1. The Donor opens the Available Requests section. 2. The system retrieves active blood requests. 3. The system filters or presents requests according to blood type compatibility and the configured proximity search. 4. The system displays available request details, including blood type, urgency, hospital, and distance when available. 5. The Donor selects a request to view its details.
Alternative or Exception Flow	2a. No active or matching requests are found: The system displays a message that no available requests were found. 2b. Request information cannot be retrieved: The system displays an error and provides a retry option. 3a. Location information is unavailable: The system indicates that proximity-based results may be limited or asks the Donor to provide location access.
Postconditions	Success: The available request list or selected request details are displayed. If no results exist, the system displays an empty-state message without changing any request.

Field	Description

Use Case ID and Name	UC-005 — View Health Tracker
Primary Actor	Donor
Goal	View donation eligibility information and the recommended next donation date.
Trigger	The Donor selects Health Tracker.
Preconditions	The Donor is logged in.
Main Flow	1. The Donor opens Health Tracker. 2. The system retrieves the Donor's verified donation history and applicable profile information. 3. The system determines the applicable waiting period based on the donation type and latest verified donation record. 4. The system displays the eligibility status, relevant waiting-period information, and recommended next donation date, where determinable.
Alternative or Exception Flow	2a. No verified donation record exists: The system displays the available default or undetermined status. 2b. The required information cannot be retrieved: The system displays an error and allows the Donor to retry. 3a. The waiting period has not elapsed: The system displays that the Donor is not yet eligible and shows the applicable remaining period.
Postconditions	Success: The current eligibility information available to the system is displayed. The tracker does not itself constitute final medical clearance for donation.

Field	Description
Use Case ID and Name	UC-006 — Accept/Decline Request
Primary Actor	Donor
Goal	Respond to an active blood request by accepting or declining it.
Trigger	The Donor selects an active request from the request list or opens it from a notification.
Preconditions	The Donor is logged in and the request is active. To accept, the Donor must satisfy the system's recorded eligibility rules and acknowledge the displayed guidelines.
Main Flow	1. The Donor opens the request details. 2. The system displays the request information and applicable eligibility guidelines. 3. The Donor chooses Accept or Decline. 4. If Accept is selected, the Donor confirms that they have read and acknowledged the displayed guidelines.| 6.The system shall send a consent form for the users to read before validating and sharing their information before being accessed by the hospital. 6. The system validates the response and rechecks that the request is still active. 7. The system records the Donor's response and acknowledgment and shows the name of the Recipient and hospital, including its date and time when accepted and routes both parties to the hospital for final eligibility verification . 8. The system updates the request response status and informs the relevant parties according to the request workflow.

Alternative or Exception Flow	3a. The Donor selects Decline: The system records the decline and leaves the request available to other potential donors, if it remains active. 4a. The Donor does not acknowledge the guidelines: The system prevents acceptance and asks the Donor to acknowledge them. 5a. The Donor does not meet a system-recorded waiting-period rule: The system blocks acceptance and displays the eligibility information. 5b. The request has been fulfilled, cancelled, or expired: The system does not record an acceptance and informs the Donor. 4b. The Donor cancels before confirmation: The system makes no change to the response.
Postconditions	Success: The response is recorded; an accepted response includes the required acknowledgment record. Decline: The decline is recorded and the request may remain open. Failed or cancelled acceptance: No acceptance is recorded.

Field	Description
Use Case ID and Name	UC-007 — Submit Blood Request
Primary Actor	Recipient
Goal	Submit a blood request so the system can search for compatible donors.
Trigger	The Recipient selects the blood request function.
Preconditions	The Recipient is logged in and has the information needed to complete the request.
Main Flow	1. The Recipient opens the blood request form. 2. The Recipient enters the required blood type, quantity, urgency, hospital location, and other requested details. 3. The system validates the request information. 4. The Recipient submits the request. 5. The system saves the request and initiates donor matching using the configured compatibility and proximity rules. 6. The system displays a submission confirmation and the current request status.
Alternative or Exception Flow	2a. Required information is missing or invalid: The system identifies the issue and asks the Recipient to correct it. 3a. The quantity or another value is outside the accepted format: The system displays a validation message. 4a. The internet connection is unavailable: The system reports that the request was not submitted. 5a. No compatible eligible donors are found: The system keeps the request active, where applicable, and informs the Recipient that no match is currently available.
Postconditions	Success: The request is saved and matching is initiated. If no donor is found, the request remains active where applicable. Failed submission: No new request is created.

Field	Description
Use Case ID and Name	UC-008 — Track Donation History
Primary Actor	Donor

Goal	Review the Donor's recorded donation history.
Trigger	The Donor opens the Donation History section.
Preconditions	The Donor is logged in.
Main Flow	1. The Donor opens Donation History. 2. The system retrieves the Donor's verified donation records. 3. The system displays the records, including available dates, locations, hospitals or blood banks, and donation status. 4. The Donor selects a record to view its details, if available.
Alternative or Exception Flow	2a. No completed donation records exist: The system displays a message indicating that no donation history is available. 2b. The records cannot be retrieved: The system displays an error and allows the Donor to retry.
Postconditions	Success: The Donor's available donation history is displayed. No donation records are changed.

Field	Description
Use Case ID and Name	UC-009 — Receive Emergency Notifications
Primary Actor	Donor
Goal	Receive an alert when a compatible and eligible match is found for an emergency blood request.
Trigger	The system identifies the Donor as a compatible and eligible match for an emergency request.
Preconditions	The Donor has an account with usable contact information. The request is active, and the Donor satisfies the configured matching criteria.
Main Flow	1. The system detects or receives an emergency blood request. 2. The matching process identifies nearby Donors who meet the applicable compatibility and eligibility criteria. 3. The system sends an emergency notification to matched Donors through the configured email channels. 4. The Donor opens the notification or the Dugo+ application. 5. The system displays the related request details and current status.
Alternative or Exception Flow	2a. No compatible and eligible Donor is found: The system sends no match notification and the request remains subject to its active-request workflow. 3a. email delivery fails: The system retains the request as active and retries notification delivery according to its retry settings. 4a. The request is no longer active: The system displays the current status rather than treating the request as available.
Postconditions	Success: A notification is attempted for each identified match and the request remains traceable. Notification failure does not by itself cancel the request.

Field	Description

Use Case ID and Name	UC-010 — View Points/Rewards
Primary Actor	Donor
Goal	View reward points earned from completed donations.
Trigger	The Donor selects the Points or Rewards section.
Preconditions	The Donor is logged in.
Main Flow	1. The Donor opens the Points or Rewards section. 2. The system retrieves the Donor's points balance and recorded reward information. 3. The system displays the current points balance and available reward information. 4. The Donor selects a reward, if available, to view its details.
Alternative or Exception Flow	2a. The Donor has no points: The system displays a zero balance. 2b. Reward information cannot be retrieved: The system displays an error and allows the Donor to retry. 4a. A selected reward is unavailable or its requirements are not met: The system displays the applicable status or requirements.
Postconditions	Success: The current points balance and available reward information are displayed. Viewing rewards does not alter the points balance.

Field	Description
Use Case ID and Name	UC-011 — View Total Units Collected
Primary Actor	Hospital
Goal	View the total number of blood units collected over a selected period.
Trigger	Hospital staff select View Total Units Collected on the dashboard.
Preconditions	Hospital staff are logged in and collection records are available, if any.
Main Flow	1. Hospital staff open the hospital dashboard. 2. Hospital staff select View Total Units Collected and specify a period, if the feature requests one. 3. The system retrieves the relevant completed-donation or collection records. 4. The system calculates the total units recorded for the selected period. 5. The system displays the total, grouped by blood type where supported.
Alternative or Exception Flow	2a. The selected period is invalid: The system asks Hospital staff to correct the period. 3a. No units have been recorded for the period: The system displays a zero total or an appropriate no-records message. 3b. The records cannot be retrieved: The system displays an error and allows Hospital staff to retry.
Postconditions	Success: The total units collected for the selected period are displayed. No collection records are modified.

Field	Description

Use Case ID and Name	UC-012 — Create Blood Requests
Primary Actor	Hospital
Goal	Create a blood request on behalf of a patient and initiate donor matching.
Trigger	Hospital staff select Create Blood Request.
Preconditions	Hospital staff are logged in and have the required patient blood-request information.
Main Flow	1. Hospital staff select Create Blood Request. 2. The system displays the request form. 3. Hospital staff enter the required blood type, quantity, urgency, hospital location, and other required request details. 4. Hospital staff submit the form. 5. The system validates the entered information. 6. The system saves the request as active and initiates matching against compatible, eligible nearby Donors. 7. The system displays the request status and sends the configured notifications to matched Donors.
Alternative or Exception Flow	3a. Required information is missing or invalid: The system identifies the fields that need correction. 5a. Validation fails: The system does not save the request and displays an error. 6a. No compatible and eligible Donors are found: The system informs Hospital staff and retains the request as active where applicable. 7a. A notification service is unavailable: The system reports the delivery issue and follows its retry process.
Postconditions	Success: The hospital request is saved and matching is initiated. Failure: The request is not saved if validation fails.

Field	Description
Use Case ID and Name	UC-013 — View Active Requests
Primary Actor	Hospital
Goal	Review active blood requests and monitor their current status.
Trigger	Hospital staff select Active Requests on the dashboard.
Preconditions	Hospital staff are logged in.
Main Flow	1. Hospital staff open Active Requests. 2. The system retrieves active requests and prioritizes them by urgency associated with the hospital or facility. 3. The system displays the requests with available details such as blood type, urgency, submission information, and current status. 4. Hospital staff select a request to view its details.
Alternative or Exception Flow	2a. No active requests exist: The system displays a no-active-requests message. 2b. The system cannot retrieve request information: The system displays an error and allows Hospital staff to retry.

Postconditions	Success: The active request list or selected request details are displayed. No request status is changed by viewing.

Field	Description
Use Case ID and Name	UC-014 — View Blood Donation History
Primary Actor	Hospital
Goal	Review a Donor's recorded donation history for operational and eligibility reference.
Trigger	Hospital staff select a Donor record and request its donation history.
Preconditions	Hospital staff are logged in, the Donor has a registered profile, and the required access authorization and consent are present.
Main Flow	1. Hospital staff locate and select a Donor record. 2. Hospital staff select View Blood Donation History. 3. The system checks the staff member's access rights and the applicable consent. 4. The system retrieves the Donor's recorded donation history. 5. The system displays the donation records available to the authorized staff member.
Alternative or Exception Flow	3a. The staff member is not authorized or the required consent is absent: The system denies access and does not display the protected history. 4a. No donation records exist: The system displays a no-history message. 4b. The history cannot be retrieved: The system displays an error and allows an authorized user to retry.
Postconditions	Success: Authorized Hospital staff can view the available history. Denied or failed access does not expose the protected records.

Field	Description
Use Case ID and Name	UC-015 — Manage Subscription Plan
Primary Actor	Hospital
Goal	View and manage the hospital's subscription plan for Dugo+ access.
Trigger	Hospital staff open Subscription Plan from the hospital dashboard.
Preconditions	Hospital staff are logged in with permission to manage the hospital subscription.
Main Flow	1. Hospital staff open Subscription Plan. 2. The system displays the hospital's current subscription information. 3. Hospital staff select an available plan or choose to retain the current plan. 4. If a change is selected, the system validates the requested plan change. 5. The system saves the plan change when valid. 6. The system displays the resulting subscription status.
Alternative or Exception Flow	3a. Hospital staff cancel the change: The system retains the current plan. 4a. The requested plan is invalid or unavailable: The system displays an error and does not apply the change. 5a. The change cannot be saved: The system reports the error and retains the previous

	plan.
Postconditions	Success: The selected valid plan is recorded, or the current plan is retained. This use case does not assume an online payment function.

Field	Description
Use Case ID and Name	UC-016 — Secure Donor Information
Primary Actor	Hospital
Goal	Access donor information only when authorized and when applicable consent is recorded.
Trigger	Hospital staff request access to a Donor's protected information.
Preconditions	Hospital staff are logged in; the Donor information exists; and applicable access authorization and consent requirements can be checked.
Main Flow	1. Hospital staff request a Donor's information. 2. The system verifies the staff member's identity and role-based access rights. 3. The system checks whether the required Donor consent is recorded. 4. The system retrieves only the information the staff member is authorized to access. 5. The system displays the permitted information and records the access event in the audit trail.
Alternative or Exception Flow	2a. The staff member is not authorized: The system denies access. 3a. Required consent is not recorded: The system does not disclose the Donor's protected information. 4a. The information cannot be retrieved: The system displays an error without exposing unauthorized data.
Postconditions	Success: Only authorized and consent-permitted information is displayed, and the access event is auditable. Failure: Protected information is not disclosed.

Field	Description
Use Case ID and Name	UC-017 — Check Blood Inventory
Primary Actor	Hospital
Goal	View current blood inventory by blood type and other recorded stock details.
Trigger	Hospital staff select Blood Inventory on the dashboard.
Preconditions	Hospital staff are logged in.
Main Flow	1. Hospital staff open Blood Inventory. 2. The system retrieves the facility's recorded inventory. 3. The system displays the available stock by blood type and recorded details such as quantity and expiration date.
Alternative or Exception Flow	2a. No inventory records exist: The system displays an empty-inventory message. 2b. Inventory information cannot be retrieved: The system displays an error and allows

	Hospital staff to retry.
Postconditions	Success: The current recorded inventory is displayed. Viewing inventory does not change stock records.

Field	Description
Use Case ID and Name	UC-018 — Edit Hospital Profile
Primary Actor	Hospital
Goal	Update the hospital's profile information.
Trigger	Hospital staff select Edit Hospital Profile.
Preconditions	Hospital staff are logged in and the hospital has an existing profile.
Main Flow	1. Hospital staff select Edit Hospital Profile. 2. The system displays the current hospital profile. 3. Hospital staff edit the permitted profile details, such as facility name, contact information, and location. 4. Hospital staff submit the changes. 5. The system validates and saves the updated information. 6. The system displays the updated profile and a confirmation message.
Alternative or Exception Flow	3a. Hospital staff cancel the edit: The system discards unsaved changes. 5a. Required information is missing or invalid: The system identifies the problem and requests correction. 5b. The changes cannot be saved: The system displays an error and retains the previous profile.
Postconditions	Success: The hospital profile is updated. Failure or cancellation: The previously saved profile remains unchanged.

Field	Description
Use Case ID and Name	UC-019 — View Nearby Hospital on Map
Primary Actor	Donor
Goal	View nearby hospitals on a map to identify a facility associated with blood donation.
Trigger	The Donor selects the nearby-hospital map function.
Preconditions	The Donor is logged in and hospital location data is available. Location permission may be needed for proximity-based display.
Main Flow	1. The Donor opens the nearby-hospital map. 2. The system retrieves available hospital locations. 3. The system uses the Donor's location, when available, to display nearby hospitals within the applicable search area. 4. The system displays the hospitals on a map. 5. The Donor selects a hospital marker to view its available details.

Alternative or Exception Flow	2a. No hospital locations are available: The system displays an appropriate message. 3a. The Donor's location is unavailable: The system indicates that nearby sorting or filtering may be limited. 4a. The map service is unavailable: The system displays an error and allows the Donor to retry.
Postconditions	Success: Available hospital locations are displayed on the map. No hospital records are modified.

Field	Description
Use Case ID and Name	UC-020 — Submit Emergency Blood Donation Requests
Primary Actor	Hospital
Goal	Submit an urgent blood request and initiate matching and emergency notifications.
Trigger	A Recipient or Hospital staff member identifies an emergency need and selects the emergency request function.
Preconditions	The actor is authenticated; the required blood type and request information are available; Hospital staff use an authorized hospital account.
Main Flow	1. The actor opens the emergency blood request form. 2. The actor enters the required blood type, quantity, urgency, hospital location, and other required information. 3. The system validates the information and identifies the request as an emergency. 4. The actor submits the request. 5. The system saves the valid request as active. 6. The system searches for compatible and eligible Donors using the configured tiered proximity radius. 7. The system sends emergency notifications to matched Donors through email and displays the request status to the submitting actor.
Alternative or Exception Flow	2a. Required information is missing or invalid: The system asks the actor to correct the form. 3a. The request fails validation: The system does not save it and displays an error. 6a. No eligible match is found within the configured search range: The system informs the actor and retains the request as active where applicable. 7a. email delivery fails: The system retains the active request and retries notification delivery according to its settings.
Postconditions	Success: The emergency request is saved and matching is initiated. If no match is found, the request remains active where applicable. Failed validation results in no new request.

Field	Description
Use Case ID and Name	UC-021 — Logout
Primary Actor	Donor, Recipient, Hospital, System Admin
Goal	Securely end the current Dugo+ session.
Trigger	The user selects Logout.

Preconditions	The user has an active authenticated session.
Main Flow	1. The user selects Logout. 2. The system invalidates the active session. 3. The system returns the user to the login screen.
Alternative or Exception Flow	2a. The session has already expired: The system returns the user to the login screen and does not retain an active session. 2b. The logout request cannot be completed because of a connection issue: The system informs the user; the session must not be treated as successfully ended until invalidation is confirmed.
Postconditions	Success: The active session is ended and the login screen is displayed. If the session had already expired, the user remains unauthenticated.

Field	Description
Use Case ID and Name	UC-022 — View Matched Party's Location
Primary Actor	Donor, Recipient
Goal	View the location of the relevant matched party after a match is established.
Trigger	The Donor or Recipient opens a matched request and selects the location option.
Preconditions	The user is logged in, a match exists, and location information is available for disclosure under the system's consent and access rules.
Main Flow	1. The user opens the matched request details. 2. The user selects View Matched Party's Location. 3. The system verifies that a match exists and that the user is permitted to view the location. 4. The system retrieves the permitted location information. 5. The system displays the matched party's location.
Alternative or Exception Flow	3a. No match exists or the user is not authorized: The system denies the location request. 4a. Location information is unavailable: The system informs the user that the location cannot be displayed. 5a. The map or location service is unavailable: The system displays an error and allows a retry.
Postconditions	Success: The authorized user views the available matched-party location. Failure: No restricted location information is disclosed.

Field	Description
Use Case ID and Name	UC-023 — View and Acknowledge Eligibility Guidelines
Primary Actor	Donor
Goal	Review the donation eligibility guidelines applicable to a selected donation type and acknowledge having read them.

Trigger	The Donor opens the eligibility guidelines before accepting a blood request.
Preconditions	The Donor is logged in and a donation type or applicable request has been selected.
Main Flow	1. The Donor opens the eligibility guidelines. 2. The system displays the guidelines relevant to the selected donation type, including applicable waiting periods and relevant eligibility considerations. 3. The Donor reviews the displayed guidelines. 4. The Donor checks the acknowledgment box to confirm that the guidelines have been read and that they meet the displayed criteria. 5. The system records the acknowledgement and its date and time. 6. The system makes the acknowledgment available to the related request workflow.
Alternative or Exception Flow	2a. The guidelines cannot be loaded: The system displays an error and does not treat the guidelines as acknowledged. 4a. The Donor does not check the acknowledgment box: The system does not record acknowledgment and prevents acceptance where acknowledgment is required. 5a. The acknowledgment cannot be saved: The system displays an error and does not proceed as though it were recorded.
Postconditions	Success: The acknowledgment and its timestamp are recorded. This acknowledgment does not replace the hospital's physical screening, tests, or medical-history verification.

Field	Description
Use Case ID and Name	UC-024 — Reset Password / Account Recovery
Primary Actor	Donor, Recipient, Hospital
Goal	Recover access to an existing account by resetting its password through a verified channel.
Trigger	The user selects Forgot Password or Account Recovery from the login page.
Preconditions	The user has an existing Donor, Recipient, or Hospital account and access to its registered verification channel. System Admin accounts are excluded.
Main Flow	1. The user opens the password recovery function. 2. The user provides the account identifier requested by the system. 3. The system verifies that the account is eligible for recovery and initiates verification through a registered channel, such as email. 4. The user completes the verification step. 5. The system allows the user to enter a new password. 6. The system validates the new password against the password requirements. 7. The system saves the new password securely and confirms that recovery is complete.
Alternative or Exception Flow	2a. The account cannot be identified: The system displays a generic recovery message. 3a. The verification message cannot be delivered: The system informs the user and offers a retry when available. 4a. Verification fails or expires: The system rejects the attempt and requests a new verification step. 6a. The new password does not meet requirements: The system explains the requirements and requests another password. 7a. The password cannot be updated: The system displays an error and does not report recovery as successful.

Postconditions	Success: The account password is updated and the user can use the new password. Failure: The existing password remains unchanged. System Admin recovery is outside this use case.

Field	Description
Use Case ID and Name	UC-025 — Request Data Access/Correction/Deletion
Primary Actor	Donor, Recipient
Goal	Submit a request to access, correct, or delete personal data held by Dugo+.
Trigger	The user selects the personal data request function from their account.
Preconditions	The user is logged in and can identify the personal data request type.
Main Flow	1. The user opens the personal data request function. 2. The system displays the available request types: access, correction, or deletion. 3. The user selects a request type and provides the required details. 4. The system validates the request information. 5. The user submits the request. 6. The system records the request and provides a reference or status, where supported.
Alternative or Exception Flow	3a. Required details are missing: The system asks the user to complete them. 4a. The request information is invalid: The system displays an error and requests correction. 5a. The request cannot be submitted because of a connection or service issue: The system informs the user and does not indicate that the request was received.
Postconditions	Success: A data-subject request is recorded for review and processing. Submission alone does not mean that the data has already been changed or deleted.

Field	Description
Use Case ID and Name	UC-026 — Update Blood Inventory
Primary Actor	Hospital
Goal	Add or update blood inventory records after a completed donation or issued transfusion.
Trigger	Hospital staff select the inventory update function following a stock-affecting event.
Preconditions	Hospital staff are logged in and authorized to update the hospital's inventory. The relevant donation or transfusion information is available.
Main Flow	1. Hospital staff open the inventory management function. 2. Hospital staff select an existing inventory record or choose to add a record. 3. Hospital staff enter or update the relevant information, such as blood type, quantity, and expiration details, as applicable. 4. The system validates the inventory information. 5. Hospital staff submit the update. 6. The system saves the inventory change and commits the corresponding audit record. 7. The system displays the updated inventory.

Alternative or Exception Flow	3a. Required information is missing or invalid: The system identifies the fields that need correction. 4a. The update fails validation: The system does not apply the change. 6a. The inventory update or audit-record operation fails: The system reports the failure and does not display the change as successfully committed.
Postconditions	Success: The inventory record is updated and the required audit event is committed. Failure: The previous inventory state remains in effect.

Field	Description
Use Case ID and Name	UC-027 — Verify Donor Eligibility
Primary Actor	Hospital
Goal	Record the result of the hospital's final donor eligibility screening.
Trigger	Hospital staff begin the eligibility verification for a matched Donor who reports to the facility.
Preconditions	Hospital staff are logged in; a matched Donor is associated with the request; and the Donor is available for screening.
Main Flow	1. Hospital staff open the matched Donor's request or record. 2. The system displays the available Donor and requests information permitted to the staff member. 3. Hospital staff conduct the required physical screening, blood tests, and medical-history check outside or alongside the system. 4. Hospital staff determine the screening result. 5. Hospital staff select and submit Cleared, Deferred, or Rejected as the Donor's status 6.Hospital confirms the donation is completed; system updates donation history, awards points, and commits the audit record. 7. The system records the screening status and updates the associated request. 8. The system makes the result available to authorized users.
Alternative or Exception Flow	2a. The Donor or request record cannot be found: The system informs Hospital staff and prevents an unrelated status update. 3a. Screening is incomplete: Hospital staff do not submit a final status until the screening decision is available. 5a. An invalid status or incomplete submission is provided: The system requests correction. 6a. The status update cannot be saved: The system displays an error and retains the previous recorded status.
Postconditions	Success: The screening result is recorded as Cleared, Deferred, or Rejected. A donation history entry is updated as completed only after Hospital staff confirm that the donation was completed.

Field	Description
Use Case ID and Name	UC-028 — Configure System Parameters
Primary Actor	System Admin
Goal	Configure system-wide matching and notification parameters.

Trigger	System Admin opens the system parameter configuration function.
Preconditions	System Admin is logged in with permission to manage system configuration.
Main Flow	1. System Admin opens System Parameters. 2. The system retrieves and displays the current configurable values. 3. System Admin edits permitted settings, including the tiered matching radius and notification retry settings. 4. The system validates the proposed values and their allowed ranges. 5. System Admin submits the changes. 6. The system saves the valid configuration. 7. The system displays a confirmation and the updated settings.
Alternative or Exception Flow	3a. System Admin cancels the changes: The system discards unsaved edits. 4a. A value is missing, invalid, or outside the permitted range: The system identifies the setting and requests correction. 6a. The configuration cannot be saved: The system displays an error and retains the previous settings.
Postconditions	Success: The valid configuration is saved for system use. Failure or cancellation: Existing configuration remains unchanged.

Field	Description
Use Case ID and Name	UC-029 — Manage User Accounts
Primary Actor	System Admin
Goal	Create Hospital accounts and suspend or remove Hospital, Donor, or Recipient accounts as needed.
Trigger	System Admin opens the user account management function.
Preconditions	System Admin is logged in and authorized to manage accounts.
Main Flow	1. System Admin opens Manage User Accounts. 2. The system displays the account management options and relevant account records. 3. System Admin selects an operation: create a Hospital account, suspend an account, or remove an account. 4. For account creation, System Admin enters the required Hospital account details. For suspension or removal, System Admin selects the target account and provides the required action. 5. The system validates the input and checks the requested operation. 6. System Admin confirms the operation. 7.The system applies the authorized change and displays the resulting account status. 8.The system now has its premade credentials for a hospital
Alternative or Exception Flow	3a. Admin attempts to create a Donor/Recipient account: system refuses; those are self-registered only 4a. Required account information is missing or invalid: The system requests correction. 5a. The target account does not exist or the operation is not permitted: The system rejects the operation and displays an error. 6a. System Admin cancels: The system makes no change. 7a. The change cannot be saved: The system displays an error and retains the previous account state.
Postconditions	Success: The authorized account is created, suspended, or removed as selected.

	Failure or cancellation: No account change is applied.

Field	Description
Use Case ID and Name	UC-030 — Process Data Subject Request
Primary Actor	System Admin
Goal	Review and process a user's request to access, correct, or delete personal data.
Trigger	System Admin opens a submitted personal data request for processing.
Preconditions	System Admin is logged in and authorized to process data-subject requests. A request has been submitted and recorded.
Main Flow	1. System Admin opens the data-subject request queue. 2. The system displays the submitted requests and their current status. 3. System Admin selects a request and reviews its type and details. 4. System Admin verifies the request and determines the appropriate action in accordance with applicable requirements. 5. System Admin records the processing decision and performs or coordinates the approved access, correction, or deletion action. 6. The system updates the request status and records the processing event. 7. The system displays the updated request status. 8. The system then commit an audit record when Admin accesses personal data.
Alternative or Exception Flow	3a. The request is incomplete or requires clarification: System Admin records the need for additional information and leaves the request pending. 4a. The request cannot be approved as submitted: System Admin records the decision and applicable reason. 5a. The requested action cannot be completed: The system records the issue and does not mark the request as completed. 6a. The status or processing event cannot be saved: The system displays an error and retains the previous recorded status.
Postconditions	Success: The processing decision and resulting status are recorded. A request is marked completed only when its approved action has been completed. Blockchain audit records remain immutable; corrections are recorded as linked entries.

Field	Description
Use Case ID and Name	UC-031 — View Audit Log
Primary Actor	Hospital, System Admin
Goal	Review authorized blockchain-based audit records for donation, inventory, and data-access events.
Trigger	An authorized user selects View Audit Log.
Preconditions	The user is logged in and has permission to view the audit records.
Main Flow	1. The user opens View Audit Log. 2. The system verifies the user's role and audit-log access rights. 3. The system retrieves the audit records the user is authorized to view.

	4. The system displays the available records, including event type, timestamp, and non-identifying references where applicable. 5. The user selects a record to inspect its available details.
Alternative or Exception Flow	2a. The user is not authorized: The system denies access. 3a. No audit records are available: The system displays an appropriate message. 3b. Audit records cannot be retrieved: The system displays an error and allows an authorized user to retry.
Postconditions	Success: Authorized audit records are displayed without modifying them. Unauthorized users cannot view protected audit information.

Field	Description
Use Case ID and Name	UC-032 — Verify Audit Record Integrity
Primary Actor	System Admin
Goal	Check whether a blockchain audit record has been altered by verifying its hash against the chain.
Trigger	System Admin selects an audit record and requests integrity verification.
Preconditions	System Admin is logged in and authorized to verify audit records; the target audit record exists.
Main Flow	1. System Admin opens the audit log. 2. System Admin selects an audit record and chooses Verify Integrity. 3. The system retrieves the record's stored hash and the corresponding blockchain data. 4. The system performs the integrity check by comparing the record hash with the value represented in the chain. 5. The system displays the verification result.
Alternative or Exception Flow	2a. The selected record cannot be found: The system informs System Admin and does not perform the check. 3a. The blockchain data is unavailable: The system reports that integrity could not be verified. 4a. The hash does not match: The system reports an integrity discrepancy and preserves the existing audit record.
Postconditions	Success: The verification result is displayed as matching or as a discrepancy. The audit record is not modified by verification.

Field	Description
Use Case ID and Name	UC-033 — View Profile
Primary Actor	System Admin

Goal	View the System Admin's own account profile.
Trigger	System Admin selects View Profile.
Preconditions	System Admin is logged in.
Main Flow	1. System Admin opens the profile function. 2. The system retrieves the authenticated administrator's profile. 3. The system displays the available profile information.
Alternative or Exception Flow	2a. The profile cannot be retrieved: The system displays an error and allows System Admin to retry.
Postconditions	Success: The administrator's own profile is displayed. No profile information is changed.

Field	Description
Use Case ID and Name	UC-034 — View Hospital Location on Map
Primary Actor	Hospital
Goal	View the hospital's saved location on a map.
Trigger	Hospital staff select the hospital location map function.
Preconditions	Hospital staff are logged in and the hospital profile contains location information.
Main Flow	1. Hospital staff open the hospital location map. 2. The system retrieves the location saved in the hospital profile. 3. The system displays the hospital's location on a map.
Alternative or Exception Flow	2a. The hospital profile has no location: The system informs Hospital staff that the location is unavailable. 3a. The map service cannot load: The system displays an error and allows Hospital staff to retry.
Postconditions	Success: The hospital's recorded location is displayed on the map. No profile or location data is changed.

2.6 System Architecture

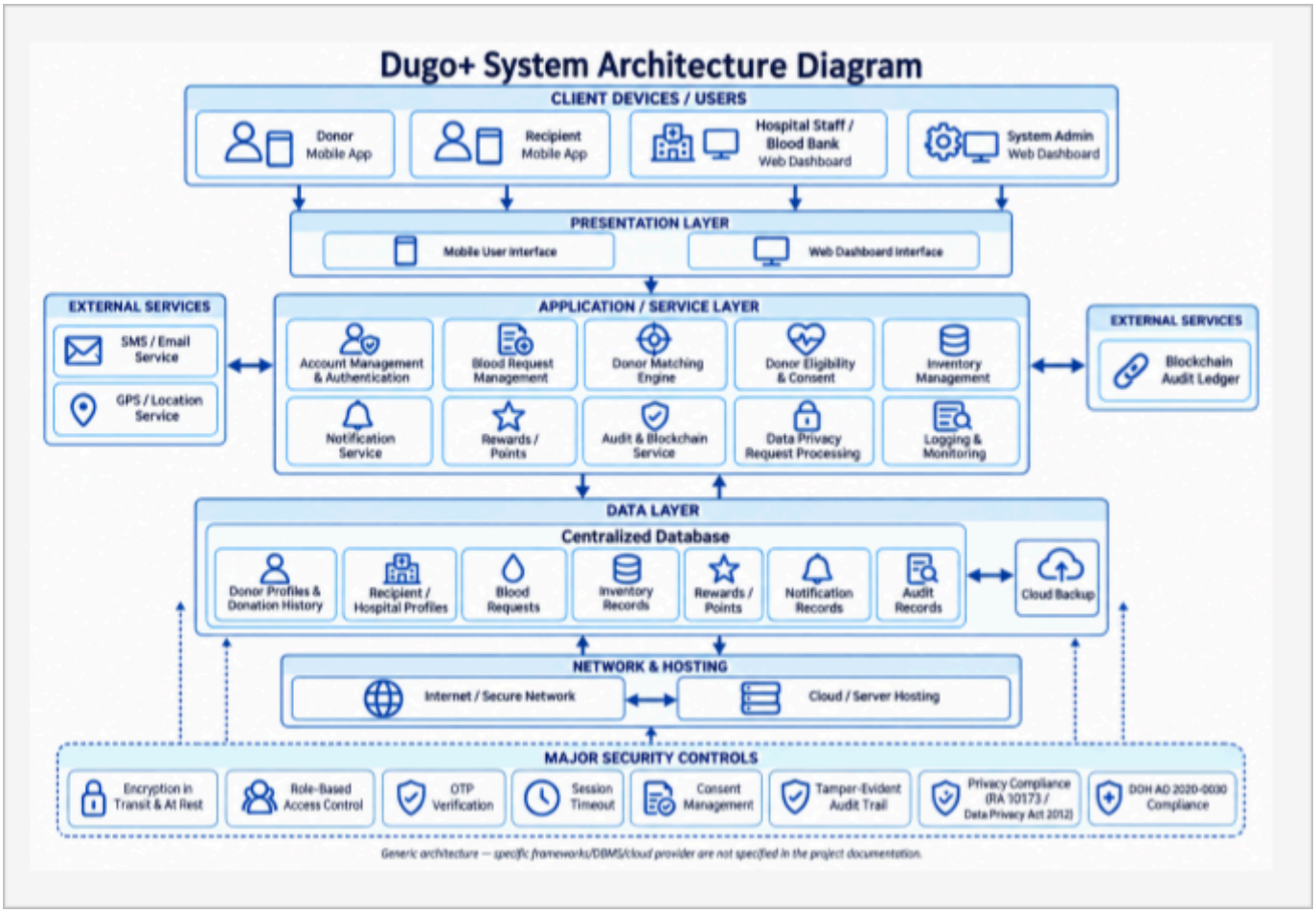


Figure 2.8. System Architecture of the Proposed System

CHAPTER 3

SYSTEM DEVELOPMENT

This chapter explains how the approved design was implemented, the methodology used, the resources required, and the user interface design.

3.1 Development Methodology

Student guidance: *Name and justify the methodology used, such as Waterfall, Agile, Scrum, Rapid Application Development, Prototyping, or an Iterative model.*

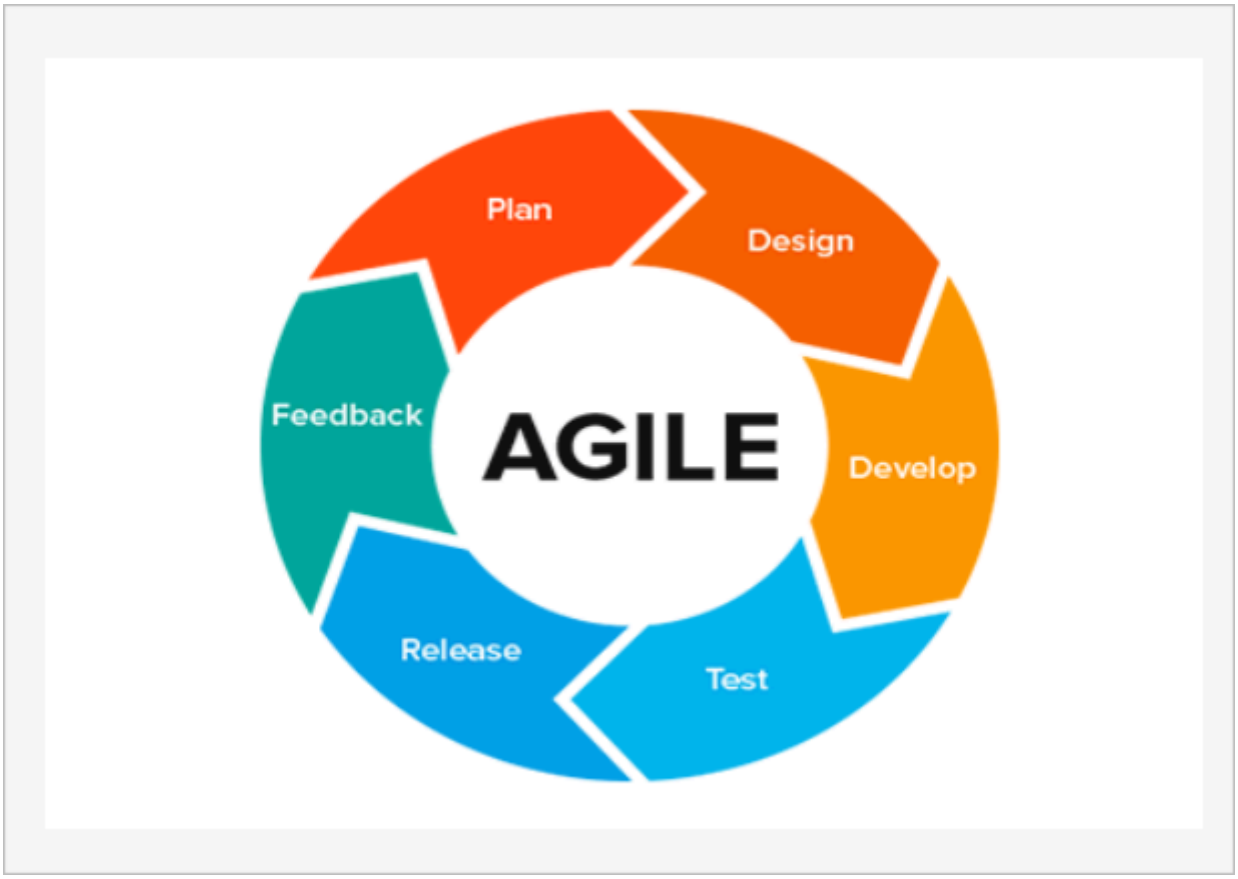


Figure 3.1. Development Methodology Applied to the Project

Explain how each phase or iteration was applied to this project, not only its textbook definition.

3.2 Software and Hardware Requirements

3.2.1 Development Requirements

Category	Item	Minimum Specification or Version	Purpose
Hardware	Developer computer	CPU + GPU: AMD Ryzen AI 9 HX 370 + Radeon 890M (16 CUs, RDNA 3.5) RAM: DDR5DDR3 > DDR5 min. 4gb ram up to 16.0 GB of ram Storage: Fast NVMe SSD (1TB+) 100GB free Processor: AMD Ryzen 5 3400G with Radeon Vega Graphics 3.70 GHz	Development and testing
Software	Operating system	Windows 11	Development environment
Software	Programming language or framework	React JS/Vite + React, Leaflet + OpenStreetMap	Application development
Software	Database management system	Firebase Cloud Firestore	Data storage
Software	Development tools	1. Figma 2. Visual Studio Code 3. Draw.io 4. Canva 5. Github & Git 6. Discord	Coding and collaboration

3.2.2 Deployment and User Requirements

User or Component	Hardware Requirement	Software or Connectivity Requirement	Notes
End user (Donor&Recipient)	1. Mobile 2.Desktop	**Mobile Browser:** Google, Safari **Desktop Browser:** Google, Safari, Microsoft Edge, Mozilla Firefox, Opera, Opera GX **Mobile OS:** IOS/Android **Desktop OS:** Windows, Linux, MacOS **Internet:** Mobile data (3G/4G/LTE minimum, 4G/5G preferred) or Wi-Fi	Minimum: 3G, enables core functions (login, view/accept requests). Recommended: 4G/LTE, 5G, or Wi-Fi required for reliable GPS-based matching and real-time map rendering (NFR-01, NFR-05).

		(minimum 5 Mbps download / 1 Mbps upload)	
End user (Hospital)	Desktop	**Desktop Browser**: Google, Safari, Microsoft Edge, Mozilla Firefox, Opera, Opera GX **Desktop OS**: Windows, Linux, MacOS **Internet:** Broadband internet (minimum 5 Mbps download / 1 Mbps upload), wired or Wi-Fi	Minimum: 5 Mbps broadband. Recommended: wired connection with backup (mobile hotspot or secondary ISP) dropped connectivity during an active emergency request is higher-risk for a hospital than for an individual user.
End user (System Admin)	Desktop	Desktop Browser: Google, Safari, Microsoft Edge, Mozilla Firefox, Opera, Opera GX Desktop OS: Windows, Linux, MacOS **Internet:** A Stable Broadband internet (minimum 5 Mbps download / 1 Mbps upload), wired or Wi-Fi	Minimum: 5 Mbps broadband. Admin actions (account management, system configuration) are infrequent but higher-impact if interrupted mid-action.
Server or hosting	[CPU, RAM, storage]	Hardware Requirement (CPU, RAM, storage): CPU: 2 vCPUs minimum, 4 vCPUs recommended RAM: 4 GB minimum, 8 GB recommended Storage: 50 GB SSD minimum, scalable/expandable as donor and donation records grow Software Requirement (OS, runtime, database): Platform: Firebase Runtime: Node.JS Database: Firebase Cloud Firestore (NoSQL) OS: Linux (e.g., Ubuntu Server LTS) Runtime: your chosen backend framework (e.g., Node.js/Express, Laravel, or Firebase Cloud Functions pick whichever your group is actually building with) Database: a relational database (e.g., MySQL or PostgreSQL) for donor, recipient, hospital, and request records; plus your blockchain-based audit ledger component (NFR-15) for immutable audit logging Since you're pairing this with a Flutter mobile web app (per your other coursework), Firebase (Firestore + Cloud Functions + Cloud Messaging for notifications) is a common, low-cost	Recommended: auto-scaling or load balancing to sustain NFR-03 (500 concurrent users) and NFR-06 (99.5% uptime).

		pairing worth considering if you haven't picked a stack yet it also gives you push notifications and hosting in one place.	
Peripheral or special device	[Device]	[Driver or integration]	N/A for current scope

3.3 User Interface Design

Student guidance: *Present the major / all screens in the order users normally encounter them. Each figure should include a readable screenshot and a brief explanation of its purpose. Protect real personal or confidential data in screenshots.*

3.3.1 Interface Screens



Figure 3.2. [Screen Name] Interface

CHAPTER 4

TESTING AND EVALUATION

This chapter provides evidence that the developed system satisfies its documented requirements and behaves correctly under expected and invalid conditions.

4.1 Testing Strategy

Student guidance: *State the testing scope, environment, testers, test data, entry and exit criteria, and treatment of defects. At minimum, cover the major functional requirements and important non-functional requirements.*

Testing Item	Project Details
Testing objective	[What the testing aims to verify]
Test environment	[Hardware, software, browser, device, network, and version]
Testers	[Developers, client representatives, intended users, or evaluators]
Test data	[Valid, invalid, boundary, and sample data used]
Entry criteria	[Conditions before testing begins]
Exit criteria	[Required pass rate or absence of critical defects]

4.2 Test Cases and Black Box Testing

Student guidance: *Black-box testing verifies system behavior from inputs and expected outputs without examining source code. Include positive cases, negative cases, boundary cases, permissions, and error handling. Each test case should trace to a requirement.*

Test Case ID	Requirement ID	Module or Feature	Precondition	Test Data or Steps	Expected Result	Actual Result	Status
TC-01	FR-01	[Feature]	[Required state]	1. [Step] 2. [Step] Data: [value]	[Observable expected result]	[Observed result]	Pass / Fail
TC-02	FR-[ ]	[Feature]	[Precondition]	[Invalid or boundary input and steps]	[Expected validation or result]	[Actual result]	[Status]
TC-03	NFR-[ ]	[Quality attribute]	[Precondition]	[Test procedure]	[Measurable criterion]	[Observed value]	[Status]

4.3 Test Summary

Module or Requirement Group	Cases Executed	Passed	Failed	Pass Rate	Remarks
[Module 1]	[No.]	[No.]	[No.]	[00.00%]	[Summary]
[Module 2]	[No.]	[No.]	[No.]	[00.00%]	[Summary]
Overall	[Total]	[Total]	[Total]	[00.00%]	[Decision]

4.4 Defect and Revision Log

Defect ID	Related Test	Description	Severity	Corrective Action	Retest Result	Status
D-01	TC-[ ]	[Observed problem]	[Critical / Major / Minor]	[Revision made]	[Pass / Fail]	[Closed / Open]
D-02	TC-[ ]	[Problem]	[Severity]	[Action]	[Result]	[Status]

CHAPTER 5

CONCLUSION AND RECOMMENDATION

This chapter summarizes what the project accomplished, states evidence-based conclusions, and proposes practical improvements or next steps.

5.1 Conclusion

Student guidance: *Answer the project objectives using evidence from the completed analysis, developed system, and test or evaluation results. Do not introduce new results. State what was achieved, what the evidence supports, and what remains subject to the stated limitations.*

[Write a concise synthesis of the problem addressed, solution developed, major verified capabilities, and overall project outcome.]

5.2 Recommendations

Student guidance: *Give specific and feasible recommendations based on the findings, limitations, failed or deferred test cases, user feedback, and future operational needs.*

REFERENCES

Student guidance: *List every source cited in the document using the citation style required by the instructor. Include policies, technical documentation, scholarly sources, and organizational documents when applicable.*

[[Author or Organization. (Year). Title. Publisher or source. URL or DOI, when applicable.]]

APPENDICES

Student guidance: *Include supporting material that is too detailed for the main chapters. Refer to every appendix in the main text and protect confidential or personally identifiable information.*

FINAL SUBMISSION CHECKLIST

- ☐ All bracketed prompts and unused sample rows have been removed.
- ☐ The objectives, requirements, diagrams, interface designs, and test cases are aligned.
- ☐ DFD levels are balanced, and all data flows are clearly named.
- ☐ The ERD includes keys, relationships, and cardinalities, and matches the data dictionary.
- ☐ Every major functional requirement has at least one test case.
- ☐ Figures and tables are readable, numbered, captioned, and discussed in the text.
- ☐ Sensitive information has been removed or anonymized.
- ☐ References and in-text citations use one consistent style.
- ☐ Include a table of contents.